

Theorem H for frozen pyrochlore ice: proofs on the cells with a coprime pair of extents and on the cells with two chains in every pair of planes, a reduction to Theorem U alone on every cell, and the cubic cells of side one to eight

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Abstract

Theorem H says that a frozen state of pyrochlore spin ice with a hexagon of odd curl has a dipole axis that is empty or carries one sign only. With Theorem P and Lemma A of an earlier note it gives the periodicity of frozen zero-flux states on every cell whose extents are all at least two. This note proves Theorem H on two families of cells and reduces it to two statements on the rest. On a cell in which two of the three extents are coprime, no sublattice of a frozen state carries all three types of vertex, and Theorem H follows in a few lines. On the cells $(2a', 2b', 2c')$ with a', b' and c' pairwise coprime, which have two chains of each family in every pair of neighbouring planes, a state is described by functions of one coordinate, and Theorem H is proved from that description and from the structure of the signed moments, which on every cell are a sum of currents along the chains when every hole is oriented. On every cell Theorem H is reduced to one statement, Theorem U, which says that in an odd state every octahedral hole has one rod pointing towards it and one away. The reduction is written out in full. Its steps are a cage law for the signed moments, the separability of line counts, a lemma on two non-negative separable functions with product zero, a law for rods on neighbouring lines with an explicit linear certificate, and the law that writes the fraction of each type on a line as the difference of two profiles of one coordinate: with every hole oriented, the count of a two-way type on a line depends on one transverse coordinate, the two sublattices share that coordinate, and with three two-way axes some type has the same count on every line of both sublattices, which the law for neighbouring lines makes one-way. An earlier form of the reduction needed a second statement, K_u , that a chain of one moment forces a direction to be absent; it is no longer a step. Theorem U is certified by DRAT proofs on the cubic cells of side four to eight and on the cells $(3, 3, 9)$, $(4, 4, 8)$, $(3, 3, 12)$ and $(6, 6, 9)$, and a state repeated periodically is a state of a larger cell, so Theorem H holds on the cubic

cells of side one to eight. A search for a counterexample to Theorem U on those cells found none. On the cubic cells of side nine and more Theorem U, and with it Theorem H, is open. The method of the two families does not carry over to three chains in every pair of planes, and the note records where it stops. The proofs were written by language-model agents and read by separate agents as checkers, with independent programs. No human has yet read them line by line.

1 The statement

The setting is that of the note on the curl reduction. The diamond lattice has sublattices A , the points with even coordinates whose sum is a multiple of four, and $B = A + (1, 1, 1)$, in units of a quarter of the cubic edge, with bonds along $d_0 = (1, 1, 1)$, $d_1 = (1, -1, -1)$, $d_2 = (-1, 1, -1)$ and $d_3 = (-1, -1, 1)$ from each A vertex. The cell (a, b, c) is the torus of a , b and c cubes along the three axes, so coordinates are taken modulo $(4a, 4b, 4c)$. Its **extents** are a , b and c , and the cubic cell of side L is (L, L, L) . The sections up to the proof of Theorem 1 are written for the cubic cell and hold on every cell with the changes of wording named in the subsection on cells that are not cubic. An ice state puts an arrow on every bond with two in and two out at every vertex, and the vertex then has a moment $n(v) = \varepsilon(v)e_{\tau(v)}$ along one of the six directions $\pm e_x, \pm e_y, \pm e_z$. The axis $\tau(v)$ is the type of the vertex and $\varepsilon(v)$ its sign. The arrows agree at the two ends of a bond exactly when

$$d_{\tau(a)} \varepsilon(a) = d_{\tau(b)} \varepsilon(b) \quad \text{on every bond } (a, b = a + d),$$

which is the **bond rule**. Two neighbours of the same type therefore have the same sign. A state is frozen when no hexagon circulates, and by Lemma L of the earlier note every hexagon of a frozen state has exactly two reversals of its flow. The hexagon has even curl when the two reversals lie on different sublattices and odd curl when they lie on the same one. A frozen state is **odd** if some hexagon has odd curl.

Say that axis k is empty in a state if no vertex has type k , one-way if the vertices of type k all have one sign, and two-way if both signs occur. All six directions occur exactly when all three axes are two-way.

Theorem H. An odd frozen state has an axis that is empty or one-way.

A chain is one of the closed $\langle 110 \rangle$ walks of bonds, of $4L$ vertices. It has an axis, the coordinate in which its two bonds at every vertex agree, and a vertex is a kink of the chain

when its type is that axis. A chain is **uniform** when all its vertices carry the same moment. By the bond rule a chain all of whose vertices have one type is uniform. In this note that type is always different from the axis of the chain, so a uniform chain has no kinks.

A **hole** is an octahedral interstice of the lattice, and the next section attaches two rods to every hole. The rods have directions given by the signs of the moments.

1.1 The statement of the reduction, and a second statement that is no longer needed

Theorem U. In an odd frozen state every hole has one rod pointing towards it and one pointing away.

Theorem H follows from Theorem U on every cell (Theorem 1 below, in its form of 30 September 2026). The first form of the reduction, of 27 September, needed a second statement as well, and the record of that statement is kept here because the certificates of the cubic cells were built on it and because its standing was corrected twice.

K_u . A frozen state that contains a uniform chain lacks one of the six directions.

Neither is proved for every cell. Theorem U is proved by hand on the cells with a coprime pair of extents and on the cells with two chains in every pair of planes, and for some holes of other cells (Theorems 10 and 11). It is certified by DRAT proofs on the cubic cells of side four to eight and on four cells that are not cubic. K_u was established on 11 September 2026 by window lemmas with DRAT certificates, one lemma proved as a strip automaton for every period, and steps proved by hand. The windows were recorded for extents of at least three along the chain and four across it. One step has a smaller scope than the record gave it. The first step says that a uniform chain makes every chain of the second family in the two neighbouring slabs of one sense. Its certificate is a window about a point of the uniform chain, and its conclusion holds only within transverse distance two of that chain, while the later steps need it on the whole slab. On a cell (a, b, c) with the chain along x the chain comes round again at transverse distance $2 \gcd(b, c)$, so the certificates give the whole slab only when $\gcd(b, c) \leq 2$. The statement on the whole slab is not a window lemma of that shape, since it is satisfiable to negate in windows of radius up to seven. It is certified on the whole torus, by DRAT proofs checked by drat-trim, for the cubic cells of side four to eight, and is open for the others. This gap was found on 27 September 2026 by an independent checker that was reading the record for another purpose.

1.2 What is proved

Theorem 1 (the reduction). On every cell, Theorem U implies Theorem H.

In its first form, of 27 September 2026, the theorem had the hypothesis that K_u holds for uniform chains of all three axes. The proof given in the section “Proof of Theorem 1” needs it no longer; the earlier proof is kept there as a second route, valid where K_u holds.

Theorem 2. Theorem H holds on the cubic cells of side four to eight, and on the cells $(3, 3, 9)$, $(4, 4, 8)$, $(3, 3, 12)$ and $(6, 6, 9)$.

Theorem 14. On a cell in which some pair of extents is coprime, every odd frozen state lacks one of the three types, so Theorem H holds.

Theorem 22. On a cell $(2a', 2b', 2c')$ with a' , b' and c' pairwise coprime, Theorem U holds.

Theorem 23. On such a cell Theorem H holds.

Theorem 24. Theorem H holds on the cubic cells of side one, two and three.

Theorems 14 and 23 are proved by hand and use neither Theorem U as a hypothesis nor K_u . Theorem 14 rests on a statement about the types of one sublattice (Theorem 12), and Theorem 23 on a description of the states of those cells by functions of one coordinate (Theorem 20) and on the structure of the signed moments (Theorems 17 and 19). Theorems 2 and 24 rest on certificates. The standing of Theorem H is then as follows.

cells	Theorem H
some pair of extents coprime, for instance $(L, L, L + 1)$	proved (Theorem 14)
$(2a', 2b', 2c')$ with a' , b' , c' pairwise coprime, for instance $(2, 2, 2)$ and $(4, 6, 10)$	proved (Theorem 23)
cubic, side one to eight; $(3, 3, 9)$, $(4, 4, 8)$, $(3, 3, 12)$, $(6, 6, 9)$ and the cells whose extents divide one of these	by certificates of Theorem U (Theorems 2 and 24, Lemma 21)
every other cell, the cubic cells of side nine and more among them	open; follows from Theorem U (Theorem 1)

On the cells of the last row Theorem H follows from Theorem U by Theorem 1, and Theorem U is open there. A pair of neighbouring planes normal to an axis holds as many chains of each family of that axis as the greatest common divisor of the two other extents. In the cells of the first row some axis has one, and in those of the second every axis has two. The method that proves Theorem H on them does not carry over to three, and a later section records where it stops. Theorem U is also proved on the cells of the first two rows (as a corollary of Theorem 12, and Theorem 22), so on those cells Theorem 1 gives a second proof of Theorem H.

Theorem P of the earlier note, that an ice state with even curl on a cell whose extents are all at least two has a one-cube period, has since been proved for every such cell in the project's working note. The hypothesis on the extents is needed: the proof makes the profile of some axis periodic with period one cube without choosing the axis, and on an axis of extent one that shift is the identity. With Lemma A for the states in which a type is absent, Theorem H gives the periodicity of frozen zero-flux states: on every cell of the first three rows of the table whose extents are all at least two, every frozen zero-flux state is invariant under a nonzero translation. On the thin cells the conclusion is false: the cells $(1, 2, 2)$, $(1, 2, 3)$, $(1, 3, 3)$, $(1, 3, 4)$ and $(1, 4, 5)$ carry frozen zero-flux states of even curl, with all six directions, that no nonzero translation fixes (solver, 1 October 2026, witnesses re-checked by an independent program and a sample run again by the main session). The cubic cell of side one has no such state (solver), so the statement holds there by computation and not through the chain. Side four is the cubic cell on which a cube-and-conquer proof of that statement had left 407 leaves unrefuted, and on which a direct certificate of H was abandoned after thirty-six hours of solving.

2 Holes, cages and rods

Put $\mathcal{H}_A = A + (2, 0, 0)$ and $\mathcal{H}_B = B + (2, 0, 0)$. These are the holes of the two sublattices. A hole O of sublattice S has six nearest vertices of S , its **methylenes** $O \pm 2e_k$, and four nearest vertices of the other sublattice S' , its **bridgeheads** $O + s$ with $s \in \{\pm 1\}^3$ and $s_x s_y s_z = -1$ for an A-hole, $+1$ for a B-hole. The ten vertices with the twelve bonds among them are the **cage** of O .

The hexagons are the pairs of holes (O, O') with $O \in \mathcal{H}_A$ and $O' = O + \sigma$, $\sigma \in \{\pm 1\}^3$,

$\sigma_x \sigma_y \sigma_z = +1$. The hexagon has the three A vertices $a_k = O + 2\sigma_k e_k$ and the three B vertices $b_k = O' - 2\sigma_k e_k$, and at a_k and at b_k it uses a chain of axis k . So a_k is a reversal of the hexagon exactly when its type is k , and likewise b_k . Writing $N(O, \sigma)$ for the number of k with $\tau(O + 2\sigma_k e_k) = k$, Lemma L reads

$$N(O, \sigma) + N(O', -\sigma) = 2 \quad \text{on every hexagon.} \quad (\text{T2})$$

Each hole lies on four hexagons, one for each admissible octant, and the b_k of a hexagon at the A-hole O are bridgeheads of O : $b_k = O + s$ where s is σ with its k -th entry reversed.

A vertex v of type k is a **rod**. It joins the two holes $v \pm 2e_k$ of its own sublattice, of which it is a methylene.

Law C. In a frozen state every hole has exactly two rods among its six methylenes.

Proof. Add (T2) over the four hexagons at an A-hole O . Each methylene $O + 2se_k$ lies in the two octants with $\sigma_k = s$, so the first terms add to twice the number of rods of O . Each bridgehead $O + s$ occurs three times among the b_k , once for each axis k , so it contributes $\sum_k [\tau(O + s) = k] = 1$, and the four bridgeheads contribute four. The right side is eight. A B-hole is the image of an A-hole under the inversion $v \mapsto (1, 1, 1) - v$, which exchanges the sublattices and preserves bonds, types and (T2). $\square$

Multiplying (T2) by σ_k before adding gives the second cage law. Write $K = [\tau = k]$.

Law P^k . For every hole O and axis k ,

$$K(O + 2e_k) - K(O - 2e_k) = \sum_s s_k K(O + s),$$

the sum being over the four bridgeheads.

Proof. On the methylene side the terms with axis $m \neq k$ cancel, because σ_k takes both signs for each value of σ_m , and the terms with axis k give $2(K(O + 2e_k) - K(O - 2e_k))$. The bridgehead $O + s$ occurs as b_k with $\sigma_k = -s_k$ and as b_m , $m \neq k$, with $\sigma_k = s_k$, so it contributes $s_k(1 - 2K(O + s))$, and $\sum_s s_k = 0$. The right side $2 \sum_s \sigma_k$ vanishes. $\square$

A hole is **straight** if its two rods lie on one axis and a **corner** otherwise.

Lemma 1. A frozen state is odd if and only if some hole is a corner. In a state with even curl the vertices of type k of each sublattice fill complete k -lines $\{v + 4ne_k\}$.

Proof. The hexagon (O, σ) is odd exactly when $N(O, \sigma)$ is 0 or 2. If it is 2, O has two

rods in one octant, which has one methylene on each axis, so O is a corner. If it is o the same holds at O' . Conversely the two rods of a corner lie in a common octant of the right parity. If every hole is straight and v has type k , the hole $v + 2e_k$ has v as a rod and hence $v + 4e_k$ as its other rod. $\square$

The rods of one sublattice thus form loops through the holes, two rod ends at every hole, and the loops of an even-curl state are straight lines round the torus.

3 Orientation and charge

A rod at $O + 2se_k$ with sign ε **points away** from O if $\varepsilon = s$ and **towards** O if $\varepsilon = -s$. The **charge** of a hole is

$$c(O) = \#\{\text{rods pointing away}\} - \#\{\text{rods pointing towards}\} \in \{-2, 0, 2\},$$

and O is **oriented** when $c(O) = 0$. Theorem U says that an odd state has no charged hole. A straight hole on axis k with rods v and $v + 4e_k$ is oriented exactly when the two rods have the same sign.

Lemma 2. Every corner of a frozen state is oriented.

Proof. The two rods of a corner O lie in an octant σ with $N(O, \sigma) = 2$, so they are the two reversals of the hexagon (O, σ) . One of them is its source and the other its sink. The two hexagon bonds at the rod $a_k = O + 2\sigma_k e_k$ both have k -component $-\sigma_k$, so a_k is a source when $\varepsilon(a_k) = -\sigma_k$, which is pointing towards O , and a sink when it points away. $\square$

So Theorem U concerns straight holes only, and it says that two consecutive vertices of type k on a k -line of an odd state have the same sign.

Write $n_k(v) = \varepsilon(v) [\tau(v) = k]$ for the k -component of the moment.

Lemma 3 (the signed cage law). In a frozen state, for every hole O and every axis k ,

$$n_k(O + 2e_k) - n_k(O - 2e_k) - \sum_s s_k n_k(O + s) = c(O).$$

So at an oriented hole the moments obey the law P^k that the types obey, and at a charged hole they fail it by the charge, by the same amount on all three axes.

Proof. The statement involves the ten vertices of one cage. Of the 3^{10} assignments of types to them, 1,200 satisfy (T2) on the four hexagons of the cage. In each the bond rule on the twelve bonds of the cage has a solution, unique up to the global reversal, which changes the sign of both sides. The identity holds in all 1,200. The restriction of a frozen state to a cage is one of these, with its signs. $\square$

The same enumeration shows that 1,104 of the patterns are oriented and 96 are not, that the 96 are all straight, and that a straight hole on axis k is charged exactly when the two bridgeheads on one side of it along k have one of the two other types and the two on the other side have the other type. This is the **unoriented cage**. Orientation is therefore a property of the types, and Theorem U is a statement about the type field alone.

4 The law for neighbouring lines

A **k -line** is a set $\{v + 4ne_k\}$, L vertices of one sublattice, indexed by its two transverse coordinates. The k -lines of A sit at even transverse positions and those of B at odd ones, and a line has four **diagonal neighbours**, the lines of the other sublattice at transverse offset $(\pm 1, \pm 1)$.

Theorem 3 (the diagonal law, R^+). Let v and w be vertices of type k on diagonally neighbouring k -lines of a frozen state, and suppose the holes of the causal diamond between them are oriented. Then $\varepsilon(v) = \varepsilon(w)$.

The causal diamond is defined, and the theorem proved, in a later section. In an odd state satisfying Theorem U the hypothesis on the holes always holds.

Theorem 4 (the line law, S). Let v and w be vertices of type k on one k -line, and suppose the holes of the line between them are oriented, together with the holes that Theorem 3 needs. Then $\varepsilon(v) = \varepsilon(w)$.

Proof. Let the holes between v and w be $O_1, \dots, O_m$. Adding Lemma 3 over them, the methylene terms telescope to $n_k(w) - n_k(v)$, and what remains is a signed sum of n_k over bridgeheads. The bridgeheads of the O_j are vertices of the four diagonal neighbours of the line, each lying strictly between v and w in the coordinate x_k and each occurring once. If one of them, u , has type k , then $\varepsilon(v) = \varepsilon(u) = \varepsilon(w)$ by Theorem 3. If none has, the sum vanishes and $n_k(w) = n_k(v)$. $\square$

5 Line counts

Fix an axis k and a sublattice S . For the k -line of S at transverse position (α, β) let $h(\alpha, \beta)$ be the number of its vertices of type k and $s(\alpha, \beta) = \sum n_k$ their signed number. A function of two variables is **separable** if it is a sum $F(\alpha) + G(\beta)$.

Theorem 5. In a frozen state h is separable. If every hole is oriented, s is separable. In general the mixed second difference of s round a hole line is, up to sign, the total charge of that line.

Proof. Let ℓ be a k -line of holes of S' at transverse position (α, β) and add P^k over its L holes. The left side telescopes to zero round the torus. The bridgehead $O + s$ lies on the k -line of S at $(\alpha + s_i, \beta + s_j)$, and as O runs over ℓ it runs over that whole line. Its coefficient s_k is $\pm s_i s_j$ with a sign fixed by the sublattice. So

$$h(\alpha+1, \beta+1) + h(\alpha-1, \beta-1) - h(\alpha+1, \beta-1) - h(\alpha-1, \beta+1) = 0.$$

The mixed second difference of h vanishes on every unit square of the lattice of lines, and adding over a rectangle gives $h(\alpha, \beta) = h(\alpha, \beta_0) + h(\alpha_0, \beta) - h(\alpha_0, \beta_0)$. With Lemma 3 in place of P^k the same sum gives the mixed second difference of s equal to $\mp \sum_{O \in \ell} c(O)$.

□

The first statement puts strong constraints on where a type can be absent. If some line of S has no vertex of type k , the lines with none form a product set $I \times J$, where I and J are the sets on which F and G take their minima.

6 Two separable functions with product zero

Lemma 4. Let $P(\alpha, \beta) = p(\alpha) + p'(\beta)$ and $Q(\alpha, \beta) = q(\alpha) + q'(\beta)$ be non-negative functions on a product $X \times Y$ with $PQ = 0$ everywhere, neither of them identically zero. Then either p and q are both constant or p' and q' are both constant.

Proof. Q is somewhere positive, so P has a zero, its minimum is zero, and its zero set is the product $Z_P = I_P \times J_P$ of the sets where p and p' are least. The same holds for Q , and $Z_P \cup Z_Q = X \times Y$. Suppose $I_P \neq X$ and $J_P \neq Y$ and take $\alpha_0 \notin I_P, \beta_0 \notin J_P$. The column $\{\alpha_0\} \times Y$ misses Z_P , so it lies in Z_Q and $J_Q = Y$. The row $X \times \{\beta_0\}$ gives

$I_Q = X$. Then Q vanishes identically. So $I_P = X$ or $J_P = Y$, and not both. If $I_P = X$, every row $\beta \notin J_P$ misses Z_P and lies in Z_Q , so $I_Q = X$. $\square$

This is the one step of the reduction that is not linear. The sheet theorem of the project's earlier work used the same fact for a single function with values 0 and 1.

Theorem 6 (stripes). Let a frozen state have every hole oriented, and suppose sublattice S has vertices of type k of both signs. Then there is an axis $n \neq k$ such that every plane $\{x_n = c\}$ of S either has no vertex of type k , or has a vertex of type k on every one of its k -lines, all of one sign.

Proof. Let P and Q count the vertices of type k of sign $+$ and $-$ on each k -line of S . Then $P = (h + s)/2$ and $Q = (h - s)/2$ are separable by Theorem 5 and non-negative. No line carries both signs, by Theorem 4, so $PQ = 0$. Neither vanishes identically. By Lemma 4 both depend on one transverse coordinate, x_n , and on each plane $x_n = c$ at most one of the two constants $P(c), Q(c)$ is positive. $\square$

7 Walls

Call a plane $\{x_n = c\}$ of one sublattice **k -free**, for $n \neq k$, if it has no vertex of type k . The planes $x_n = c$ alternate between the sublattices as c increases.

Theorem 7 (walls). Let an odd frozen state have every hole oriented and let axis k be two-way. Then there are an axis $n \neq k$ and a k -free plane $F = \{x_n = c\}$ such that the vertices of type k on each of the two neighbouring planes $x_n = c \pm 1$ have one sign, the two planes do not have the same sign, and at least one of them has a vertex of type k .

Proof. Label every plane $x_n = c$ by the set of signs of its vertices of type k .

Suppose first that some sublattice S has both signs. Theorem 6 gives the axis n , and every plane of S is labelled $\emptyset, \{+\}$ or $\{-\}$, the last two being full in the sense of the theorem. A vertex of type k on a plane of S' next to a full plane of S of sign $+$ lies on a line diagonally neighbouring a line of that plane, so it has sign $+$ by Theorem 3. If S' also has both signs it is striped, and its stripes run along n , because a stripe of sign $-$ along the other transverse axis would meet the plane of S' next to a full plane of S of sign $+$. So every plane of either sublattice has at most one sign. Going round the torus in x_n , take a plane c^* of sign $+$ such that the next plane with any vertex of type k , say c^{**} , has sign $-$. If the two were adjacent, one of them would be a full plane of S and the other would have its sign. So they are not

adjacent, and $F = \{x_n = c^* + 1\}$ has the properties claimed.

Otherwise each sublattice has at most one sign, and since the axis is two-way, A has sign σ and B has sign $-\sigma$. By Theorem 3 no line of A with a vertex of type k is a diagonal neighbour of such a line of B. The empty lines of A form a product $I \times J$ by Theorem 5, and it is not empty, since otherwise B would have no vertex of type k . If I and J were both proper, A would have a full column and a full row of lines, the two neighbouring columns and the two neighbouring rows of B would be empty, and B would be empty altogether. So the empty lines of A form complete planes, the others form full planes, and the same holds for B, with the same normal n because a full plane of A of one normal has lines neighbouring those of a full plane of B of the other. Full planes of A and B are never adjacent. Take a full plane c^* of A whose next full plane is a plane of B. Then $c^* + 1$ and $c^* + 2$ are k -free and $F = \{x_n = c^* + 1\}$ serves. $\square$

The second case has not been seen in any computed state and is unsatisfiable on the cells of side three and four. It is kept because the argument needs it.

Lemma 5. Let $F = \{x_n = c\}$ be as in Theorem 7. At every hole O of F , the two bridgeheads of O on the plane $c + 1$ are both of type k or neither is, and the same holds for the two on the plane $c - 1$.

Proof. The methylenes $O \pm 2e_k$ lie in F and are not of type k . Let $T_{\pm} = \sum s_k K(O + s)$ over the two bridgeheads with $s_n = \pm 1$. Law P^k gives $T_+ + T_- = 0$. If one of the neighbouring planes is k -free its T vanishes and so does the other. If not, the two planes have signs $\sigma_+ = -\sigma_-$, the hole is oriented, and Lemma 3 gives $\sigma_+ T_+ + \sigma_- T_- = 0$, so again both vanish. The two bridgeheads with $s_n = +1$ have opposite s_k , so $T_+ = 0$ says that their indicators agree. $\square$

The statement of Lemma 5 for an arbitrary k -free plane was the conjecture (E') of the project's reports. In that generality it is false, since a k -free plane with a hole whose upper bridgeheads disagree exists in frozen states of even curl. At the walls of an odd state it is a consequence of orientation.

8 Uniform chains beside a wall

Two facts from the earlier work are needed. The first has a short proof.

Lemma 6 (grid counts). In a frozen state the number of vertices of type j on the plane

$\{x_j = t\}$ is the same for all planes of A and the same for all planes of B.

Proof. Let $f(t)$ be that number. Add P^j over the holes of the plane $x_j = t$. Every vertex of the plane $t+2$ is $O+2e_j$ for one hole O of the plane t , and every vertex of the plane $t+1$ is a bridgehead, with $s_j = +1$, of two of them. So $f(t+2) - f(t-2) = 2(f(t+1) - f(t-1))$ for every t . With $g(t) = f(t+1) - f(t-1)$ this reads $g(t+1) + g(t-1) = 2g(t)$. A periodic sequence with vanishing second difference is constant, and $\sum_t g(t) = 0$. $\square$

Theorem M' (proved in the project's reports). On a cubic cell (and on every cell: see the subsection on cells that are not cubic), a frozen state in which one sublattice has no vertex of type k and the other has one has even curl.

Its proof, by two discrete integrations on the transverse torus and finite lemmas on four hexagons with DRAT certificates, is in the report on the sheet theorem.

Theorem 8. Let an odd frozen state have all three types, let $F = \{x_j = c\}$ be a k -free plane of A at whose holes the conclusion of Lemma 5 holds for the plane $c+1$, and suppose the plane $c+1$ has a vertex of type k . Then every vertex of type k on the planes $c+1$ and $c+2$ lies on a uniform chain, a chain of axis j all of type k .

Proof. Write the coordinates in the order (x_i, x_j, x_k) . For a hole O of F the two bridgeheads on the plane $c+1$ are $O + (1, 1, -1)$ and $O + (-1, 1, 1)$, and every pair of vertices of that plane differing by $(2, 0, -2)$ arises so. Hence K is constant on every diagonal of the plane $c+1$ in the direction $(1, 0, -1)$.

Let O' be a B-hole of the plane $c+1$. Its bridgeheads on the plane c are not of type k , and those on the plane $c+2$ are $g_1 = O' + (1, 1, 1)$ and $g_2 = O' + (-1, 1, -1)$, so P^k at O' reads $K(O' + 2e_k) - K(O' - 2e_k) = K(g_1) - K(g_2)$. A vertex g of the plane $c+2$ has two neighbours below it, $g + (1, -1, -1)$ and $g + (-1, -1, 1)$, which differ by $(2, 0, -2)$ and so have a common value $\kappa(g)$ of K . For g_1 these neighbours include $O' + 2e_k$ and for g_2 they include $O' - 2e_k$. So $\Delta(g) = K(g) - \kappa(g)$ satisfies $\Delta(g_1) = \Delta(g_2)$, and Δ is constant on every diagonal of the plane $c+2$ in the direction $(1, 0, 1)$.

If $\Delta = -1$ on such a diagonal, the vertices below it form a diagonal of the plane $c+1$ in the direction $(1, 0, 1)$, all of type k . It meets every diagonal in the direction $(1, 0, -1)$, so the whole plane $c+1$ is of type k and has no vertex of type j . By Lemma 6 B has no vertex of type j , the state has one, and Theorem M' gives even curl. So $\Delta \geq 0$, and both neighbours above a vertex of type k of the plane $c+1$ are of type k .

The plane $c + 1$ has a vertex of type k . Its diagonal in the direction $(1, 0, -1)$ is all of type k and meets every diagonal of that plane in the direction $(1, 0, 1)$. So each of those contains a vertex w' of type k , and the vertex $g = w' + (1, 1, -1)$ above it has $\kappa(g) = 1$, hence $K(g) = 1$ and $\Delta(g) = 0$. The map $w' \mapsto w' + (1, 1, -1)$ carries the diagonals of the plane $c + 1$ in the direction $(1, 0, 1)$ onto those of the plane $c + 2$, so $\Delta = 0$ on the whole plane $c + 2$. A vertex of the plane $c + 2$ is of type k exactly when its two neighbours below are, and a vertex of the plane $c + 1$ exactly when its two neighbours above are. Those neighbours are its neighbours on the chain of axis j through it in the slab of the two planes. So a chain of that slab with one vertex of type k is all of type k . $\square$

By the inversion and the reflections of the lattice the theorem holds with A and B exchanged and on the lower side of the plane.

9 The diagonal law on the causal diamond

This section proves Theorem 3. Take $k = z$ and, by the symmetries, v an A vertex at the origin and w at (δ_x, δ_y, s) with $|\delta_x| = |\delta_y| = 1$ and s odd and positive. For $s = 1$ the two are bonded and the bond rule gives the result. Suppose $\varepsilon(v) = +1$ and $\varepsilon(w) = -1$. Then v points into the hole $h_v = v + 2e_z$ from below and w points into $h_w = w - 2e_z$ from above.

Two holes joined by a hexagon differ by a vector $\tau \in \{\pm 1\}^3$. An upward path is a sequence of such steps with $\tau_z = +1$, and the **causal diamond** D is the set of holes lying on an upward path from h_v to h_w . It is an octahedron of holes with apices h_v and h_w , of transverse radius $(s - 1)/2$ at its middle.

For an oriented hole write $\text{in}(h)$ for the unit vector from h to the rod that points towards it.

Lemma 7 (conservation). If two oriented holes O and $O + \tau$ are joined by a hexagon, then $\tau \cdot \text{in}(O) = \tau \cdot \text{in}(O + \tau)$, and the same for the rods pointing away.

Proof. The two cages have fourteen vertices and seven hexagons. Of the 3^{14} assignments of types 6,192 satisfy (T2) on the seven, each has a sign field unique up to reversal, and 5,136 have both holes oriented. The relation holds in all of these. $\square$

When s is 3 or 5 the holes h_v and h_w are joined by a hexagon, and Lemma 7 contradicts the two rods. For larger s conservation along paths is not enough, and the proof is a single

linear identity over the diamond.

Introduce for every vertex u and axis k two variables $p_k(u), m_k(u) \geq 0$, to be read as the indicators that u has moment $+e_k$ or $-e_k$, and let $n(u) = \sum_k (p_k(u) - m_k(u))e_k$. A frozen state whose diamond is oriented satisfies the following linear equations, each written with its right side subtracted.

- One-hot at u : $\sum_k (p_k(u) + m_k(u)) - 1 = 0$.
- (T2) at a hexagon: $\sum (p_k(u) + m_k(u)) - 2 = 0$, the sum over its six vertices u with k the axis the hexagon uses at u .
- The bond rule at a bond $(a, a + d), a \in A$: $d \cdot n(a) - d \cdot n(a + d) = 0$.
- Orientation at a hole h of D : $\sum_k (p_k(h + 2e_k) + m_k(h - 2e_k)) - 1 = 0$.
- The hypotheses: $p_z(v) - 1 = 0$ and $m_z(w) - 1 = 0$.

For a vertex u let $\mu(u)$ be the number of holes of D of which u is a bridgehead, and for a bond let ν be the number of the six hexagons through it that have both their holes in D .

Theorem 9 (the certificate). For every odd $s \geq 5$, the sum of the orientation equations of the holes of D , minus half the (T2) equations of the hexagons with at least one hole in D , plus $\frac{1}{2} \min(2, \mu(u))$ times the one-hot equation of every vertex, plus $\frac{1}{4} \min(4, \nu)$ times the bond equation of every bond, plus the two hypotheses, is identically

$$-1 - \sum_j r_j x_j$$

in the variables x_j , with every $r_j \geq 0$.

The left side vanishes on a state satisfying the equations and the right side is at most -1 , so no such state exists, which is Theorem 3 for $s \geq 5$. The identity is a finite identity on the infinite lattice. A state on a torus lifts to a periodic state of the lattice that satisfies every lifted equation, so the theorem holds on every torus, whether or not the diamond wraps round it. Negative separations and the other sublattice order follow by the symmetries of the lattice, which permute the variables and preserve the system of equations.

Exactly four of the coefficients r_j are non-zero, each equal to 1. They belong to the variables that say that the hole of D extreme in the direction $+x, -x, +y$ or $-y$ has its inward rod on that axis on the outer side. Without the two hypotheses the identity is therefore a statement about the six corners of the octahedron.

Corollary (the corner law). In a frozen state with D oriented, exactly one of the six corner holes of D has its inward rod on its own axis, on the side away from D .

Theorem 3 says that the two apices cannot both be such corners. The same certificate, with w on the line of v , is valid and gives the line law directly.

Proof of Theorem 9. A vertex u lies in the cages of ten holes. Six are the holes $u \pm 2e_k$ of which it is a methylene and four are the holes $u + \sigma$ of which it is a bridgehead. The hole $u + 2se_k$ is joined by a hexagon to $u + \sigma$ exactly when $\sigma_k = s$, so the ten holes are the four vertices and six edges of a tetrahedron $T(u)$, and the twelve hexagons through u are its twelve incidences. Every equation in which a variable of u occurs has a multiplier determined by which holes of $T(u)$ lie in D . So the coefficient of each variable of u in the combination is a function of the set $D \cap T(u)$.

In the coordinates $t = z$, $a = x + y$ and $b = x - y$ the diamond is cut out by eight inequalities, four for the future of h_v and four for the past of h_w . Each of them cuts $T(u)$ in the whole of it, or nothing, or a vertex with its three edges, or the opposite face. The intersections of such sets number fifty, in nine classes under the symmetries of the tetrahedron, and the coefficients can be computed for each. They are zero in seven classes. In the class of a single edge, a hole $h = u - 2d$ alone, the coefficient is 1 on the variable saying that u is the inward rod of h and zero elsewhere. In the class of two faces, where u is a bridgehead of exactly two holes of D , a coefficient of the wrong sign occurs.

That class does not arise. If u were a bridgehead of exactly two holes of D , then, because the future of h_v is closed upwards and the past of h_w downwards, either both lower bridge holes of u are in D and neither upper one, or the reverse, or one of each. In each case the inequalities, written in the coordinates a and b separately, would require the numbers of steps available in the two coordinates to differ by more than one, which they never do, or by an amount that is possible only when the transverse offset of w is at least two. The case analysis is given in full in the report of the agent that found it and was derived again by the checker. At any transverse offset of two or more the class does arise, from the first admissible separation, and the linear programme is then feasible.

The variable $p_z(u)$ has a non-zero coefficient only when $u + 2e_z$ is a hole of D with no hole of D below it, which is h_v , and the hypothesis on v cancels it. To find the constant, evaluate the identity at the state in which every vertex has moment $+e_z$. It satisfies every

equation except $m_z(w) = 1$, where the left side is -1 , and the variables p_z have coefficient zero. $\square$

The finite parts of this proof are the cut structure of the tetrahedron, the table of fifty intersections, and the agreement of the local coefficients with those of the whole combination. Each was computed by two independent implementations. The identity itself was verified in exact rational arithmetic for every odd s from 5 to 101, for every admissible transverse offset and both sublattice orders, and on the line for every gap up to ten.

Orientation is used at every hole of the diamond. Removing the orientation equation of any one hole makes the system feasible, and orientation of the corner holes alone, which Lemma 2 provides without any conjecture, does not suffice. So the diagonal law as proved here depends on Theorem U.

10 Proof of Theorem 1

The proof uses Theorem 15 of the section on cells with a coprime pair of extents, which holds on every cell: on each sublattice the fractions of the three types on the three lines through a vertex are differences of three profiles of one coordinate. It was found on 30 September 2026 by combining that law with Theorem 6 across the three axes at once. The earlier proof, which needed K_u , follows it.

For an axis k and a sublattice S , the k -lines of S are indexed by their two transverse coordinates (α, β) , each of the parity of S , and every such pair is a line. Write $h_k^S(\alpha, \beta)$ for the number of vertices of type k on the line, as in Theorem 5. A plane $\{x_n = c\}$ of S , with $n \neq k$, is k -free if no line in it has a vertex of type k , and full if every line in it has one.

Lemma 22. Let a frozen state have every hole oriented and let axis k be two-way on S . Then there is an axis $n \neq k$ such that h_k^S is a function of x_n alone; every plane $\{x_n = c\}$ of S is k -free or full, its lines all carry the same number of vertices of type k , and those vertices have one sign.

Proof. This is the proof of Theorem 6 read once more. The counts $P = (h + s)/2$ and $Q = (h - s)/2$ of the two signs are separable by Theorem 5, non-negative, with $PQ = 0$ by Theorem 4, and neither is identically zero because both signs occur on S . Lemma 4 makes both functions of one coordinate x_n , so $h = P + Q$ is one too, and on each plane at

most one of $P(c)$, $Q(c)$ is positive. $\square$

Lemma 23 (the two sublattices share the normal). Let a frozen state have every hole oriented, let axis k be two-way, and let both sublattices carry type k . Then there is one axis $n \neq k$ such that h_k^A and h_k^B are both functions of x_n alone, every plane $\{x_n = c\}$ of either sublattice is k -free or full with one sign, and two adjacent non-empty planes, one of each sublattice, have the same sign.

Proof. Three cases, which exhaust the ways in which a two-way axis can be carried by two sublattices.

Both sublattices two-way. Lemma 22 gives normals n for A and m for B. Suppose $m \neq n$, so that $\{n, m\}$ are the two transverse axes. A has a full plane $x_n = c$ of sign $+$ and B a full plane $x_m = d$ of sign $-$. The B-line at $(x_n, x_m) = (c + 1, d)$ exists, carries a vertex of type k of sign $-$, and is a diagonal neighbour of the A-lines $(c, d \pm 1)$, which carry vertices of type k of sign $+$. Theorem 3 is contradicted, so $m = n$. If the B-plane $c + 1$ is not empty, each of its lines is a diagonal neighbour of lines of the A-plane c , and Theorem 3 gives it the sign of that plane.

A two-way, B of one sign σ . Lemma 22 gives n for A, and A has a plane $x_n = c$ of sign $-\sigma$. Every B-line at $x_n = c \pm 1$ is a diagonal neighbour of lines of that plane, so by Theorem 3 it has no vertex of type k . By Theorem 5 the empty lines of B form a product $I \times J$, which contains $\{c - 1, c + 1\} \times Y_B$, so $J = Y_B$; then $F(c + 1) + G(\beta) = 0$ for every β makes G constant, and h_k^B is a function of x_n whose non-empty planes are full. The case with A and B exchanged is the same.

A of one sign σ , B of one sign $-\sigma$. This is the second case of the proof of Theorem 7, which uses Theorems 3 and 5 and not oddness: the empty lines of A form complete planes normal to an axis n , the same holds for B with the same n , and full planes of A and B are never adjacent. With the empty set of the form $X_A \times J$ the separable h_k^A has its first summand constant and is a function of x_n , and likewise h_k^B . $\square$

Lemma 24 (the killed profile). Let $P(x)$, $Q(y)$ and $R(z)$ be the profiles of Theorem 15 on S , so that $h_y/\ell_y = P - R$, $h_z/\ell_z = Q - P$ and $h_x/\ell_x = 1 + R - Q$. If h_k^S is a function of x_n alone, $n \neq k$, then the profile of the third coordinate, $i(k) \notin \{k, n\}$, is constant on S .

Proof. Take $k = x$, so that $h_x/\ell_x = 1 + R(z) - Q(y)$ on all pairs (y, z) of the parity of

S , and every pair occurs. If this does not depend on z then R is constant, and if not on y then Q is. The other two axes are the same. $\square$

Lemma 25 (full lines carry one sign). Let a frozen state have every hole oriented. If every k -line of both sublattices has a vertex of type k , then axis k is one-way.

Proof. Join two k -lines when they are diagonal neighbours. The lines of both sublattices are the pairs (α, β) with $\alpha + \beta$ even, and the steps $(\pm 1, \pm 1)$ generate that set, so the graph is connected on every torus. Along an edge Theorem 3 gives equal signs to the vertices of type k on the two lines, and every line has one. $\square$

Proof of Theorem 1. Let the state be frozen and odd and assume Theorem U, so that every hole is oriented. If some type is absent its axis is empty. Otherwise all three types occur, and by Theorem M' both sublattices carry all three, since a sublattice lacking a type that the other has would make the curl even. Suppose every axis is two-way. Lemma 23 gives each axis k a normal $n(k) \neq k$, the same for A and B, and by Lemma 24 the profile of the third coordinate $i(k) \notin \{k, n(k)\}$ is constant on each sublattice. Now $i(x) \in \{y, z\}$, $i(y) \in \{x, z\}$ and $i(z) \in \{x, y\}$, so the three values $i(x), i(y), i(z)$ are not all equal. If they are all different, every profile is constant and so is every count h_k . If they take two values a and b , the profiles of a and b are constant, and the count h_c of the third axis c is ℓ_c times a difference of those two profiles plus a constant, so it is constant on each sublattice. In either case some axis c has h_c^A and h_c^B constant, and the constants are at least 1 because both sublattices carry type c . So every c -line of both sublattices has a vertex of type c , and by Lemma 25 axis c is one-way, which contradicts the supposition. Some axis is therefore not two-way, and since it is not empty it is one-way. $\square$

Theorem U enters through Theorems 3 and 4 and through the second statement of Theorem 5, for the two-way axes only. Oddness enters through Theorem U and through Theorem M' . Neither K_u , the wall of Theorem 7, Theorem 8 nor Theorem 19 is used; the third case of Lemma 23 repeats the argument of the second case of Theorem 7. On a cell with an extent of one the methylenes $O \pm 2e_k$ of a hole coincide; such a cell is covered by Lemma 21, since a periodic repetition preserves frozenness, oddness, the moments, the types carried by each sublattice, which axes are two-way and the charge of every hole, the charge being counted over the six methylene positions. Theorem M' is used with the standing the first section gives it on cells that are not cubic and at extent two.

The argument makes one prediction that can be tested on odd states whose axes are not all two-way. If the state has all three types, a one-way axis c and two two-way axes a and b , Lemmas 24 and 25 forbid the normals of a and b from making a or b the axis of constant count, so $n(a) = b$ and $n(b) = a$. In the 429 states of the corpus of the cubic cells of side three and four with this pattern the two two-way axes are each other's normals in every one, and the constant profile is the one the argument names.

10.1 The earlier proof, through K_u

Let the state be frozen and odd, on a cell on which K_u holds for uniform chains of all three axes, and assume Theorem U, so that every hole is oriented. If an axis is empty or one-way there is nothing to prove. Otherwise all three types occur and every axis is two-way. Take any axis k . Theorems 3 and 4 hold at every separation, so Theorem 7 gives a wall F with a neighbouring plane that has a vertex of type k , Lemma 5 gives the hypothesis of Theorem 8 at F on that side, and Theorem 8 gives a uniform chain. By K_u a direction is absent, which contradicts the assumption that every axis is two-way. $\square$

Oddness enters twice, through Theorem U and in Theorem 8, where Theorem M' excludes a complete plane of one type. The signs of the moments enter through Lemma 3 and the bond rule. The argument names one axis and one wall, and it does not use the trails of odd hexagons, the run laws or the statements on three planes that the project's earlier reports pursued. The wall's normal is produced by Theorem 6 and cannot be chosen, which is why K_u is needed for chains of every axis. This proof was written on 27 September 2026, before Theorem 15 was found, and the two results it combines with K_u , Theorems 6 and 15, were not put together across the three axes until 30 September.

11 Theorem H on the cubic cells of side four to eight, and on four cells that are not cubic

Theorem U follows from a statement about four vertices. By the enumeration after Lemma 3 an unoriented hole has both rods of type k , a bridgehead of type a on one side and a bridgehead of type b on the other, where a and b are the other two axes.

Theorem U' (certified for the cubic cells $L = 4$ to 8 and for $(3, 3, 9)$, $(4, 4, 8)$,

$(3, 3, 12)$ **and** $(6, 6, 9)$). A state of the type model, that is, an assignment of types satisfying (T_2) , that has a hole with both rods of type k , one bridgehead of type a on its lower side and one of type b on its upper side, has even curl on every hexagon.

The type model has no signs and is weaker than the ice rule, so the theorem proves more than Theorem U needs. Its negation was written as a formula in conjunctive normal form for each position of the two bridgeheads, each order of the two types and each sublattice, refuted by CaDiCaL with a DRAT proof, and the proof checked by drat-trim.

statement	L	formulas	longest solve	proof size	drat-trim
Theorem U'	4	16	2.4 s	36 to 45 MB	all verified
Theorem U'	5	16	9.5 s	114 to 147 MB	all verified
Theorem U'	6	4	22 s	315 to 385 MB	all verified
diagonal law	4, 5, 6	16, 20, 24	6.6 s	up to 12 MB	all verified
line law	4, 5, 6	4, 6, 8	3.8 s	up to 7 MB	all verified

At side six the four formulas are one for each sublattice and order, and the sixteen hypotheses were checked to form one orbit of the symmetries of the formula. The diagonal and line laws were certified directly on these cells, in the model with signs and without any hypothesis of oddness, before Theorem 9 was found. They are recorded because they do not depend on it. Every formula has a control, the same formula without the negated conclusion, which is satisfiable, and the witness was decoded and checked against the definitions. The encoding was tested on 1,357 odd frozen states found earlier, each of which satisfies the hypothesis part of every formula wherever the pattern occurs.

On 1 October 2026 the same statement was run on the cheapest cells on which Theorem U was open, as a search for a counterexample: a satisfiable formula would have been an odd state of the type model with an unoriented hole, to be lifted to the bond model and tested against Theorem H. The encoding was generalised to any cell (on the cubic cells of side four and five its clauses are identical to the earlier encoding's), and tested on odd corpus states repeated onto $(3, 3, 9)$ and $(4, 4, 8)$, which satisfy its hypothesis part and in which the full pattern of U' occurs nowhere. A query is a rod axis, a sublattice, an order of the two other types and a position of each bridgehead, sixteen per axis; on a cubic cell the cyclic permutation of the axes is a symmetry of the formula (checked as

a clause-set automorphism at sides seven and eight) and the sixteen hypotheses on one axis form one orbit, as at side six; on a cell that is not cubic the transposition of the two unequal axes is not a symmetry, and the forty-eight hypotheses fall into three orbits (the two bridgehead diagonals on the short axes, and the long axis), so all forty-eight were run. Every query was unsatisfiable, with a satisfiable control whose witness was decoded and checked. Certificates were then written for every query on the three small cells, for one query per orbit on $(6, 6, 9)$, and for one per sublattice and order at sides seven and eight.

cell	queries run	formulas			
		certified	longest solve	proof size	drat-trim
$(3, 3, 9)$	48	48	2.7 s	17 to 35 MB	all verified
$(3, 3, 12)$	48	48	4.4 s	18 to 50 MB	all verified
$(4, 4, 8)$	48	48	8.7 s	73 to 151 MB	all verified
$(6, 6, 9)$	48	3	35 s	753 to 900 MB	all verified
$(7, 7, 7)$	16	4	48 s	0.9 to 1.3 GB	all verified
$(8, 8, 8)$	16	4	124 s	2.0 to 2.9 GB	all verified

Two hundred and twenty-four queries, none satisfiable, none timed out; about 106 minutes of solving and checking in all. The search found no near miss: every control is satisfiable in under half a second and every query is refuted in at most about two minutes, and the solving time on the cubic cells grows smoothly with the side, 2.4, 9.5, 22, 48 and 124 seconds from side four to side eight. By Lemma 21 the certificates also give Theorem H on every cell whose extents divide one of these six, up to a permutation of the axes, among them $(3, 6, 9)$, $(4, 8, 8)$ and $(2, 8, 8)$.

Theorem 2 follows from Theorem 1 and these certificates of Theorem U' alone. The first form of Theorem 1 needed K_u on these three cells as well, and, as the first section says, K_u on a cubic cell needs its first step on the whole slab, which the certificates of 11 September did not provide. It was provided by the following certificates, which are kept as certified facts about K_u and are no longer steps of the proof of Theorem 2: on the whole torus, with the ice rule at every vertex and (T2) on every hexagon, a uniform chain and a chain of the second family in a neighbouring slab with a vertex of the other sense are contradictory.

statement	L	formulas	longest solve	proof size	drat-trim
first step of K_u , whole slab	4	2	0.2 s	0.3 to 0.5 MB	verified
	5	2	1.1 s	4 MB	verified
	6	2	6.9 s	26 to 27 MB	verified
	7	2	32 s	122 to 125 MB	verified
	8	2	164 s	704 to 734 MB	verified

The two formulas are for the slab above the chain and the slab below it, each with a satisfiable control. They are written for one uniform chain, and the others follow by the symmetries of the cubic cell. The cubic cells of side one, two and three follow from Theorem 2 by Theorem 24.

One fact about the first step of K_u was found on 30 September 2026 and is recorded although the step is no longer needed. The step is not a window lemma of bounded radius even when the hypotheses that the earlier proof of Theorem 1 has at the point of use are added (a k -free wall, one sign on the planes beside it, orientation, oddness, the uniform chains of Theorem 8): every window that does not contain the uniform chain is satisfiable. When the chain is in the window and every hole is oriented, the statement is local in a cone: a vertex of a column of a neighbouring slab at transverse distance d from the chain is forced to the sense of the chain by a window of transverse radius d about it and about d grids of extent along the axis of the chain on each side, and by nothing smaller (solver, five cells, 572 windows per extent, to $d = 7$). Orientation does the work; the wall adds nothing. This is the shape of Theorem 3 on the causal diamond, and the closed-form certificate exists: the first step of K_u at a vertex t of a neighbouring column at transverse distance $d \geq 2$ from the chain, given orientation, is the sum of two identities of Theorem 9 on two causal diamonds of separation $2d - 1$ along the axis of the chain's moment, whose lower apices are the holes above two vertices of the chain. The first diamond ends at the hole below t , so t cannot carry the moment opposite to the chain's (Theorem 3). The second ends at a hole whose corner extreme in the third direction is the single hole beside t , and its lower

apex is already the one corner that the corner law allows, so t cannot carry the moment of the third axis either. The distances 0 and 1 follow from the bond rule. So, given Theorem U, the first step of K_u holds by hand on every cell, with no dependence on the side, and with the later steps, which hold on every cell, K_u given U is a theorem on every cell (found by one agent on 30 September 2026 and checked exactly on the infinite lattice for $d = 2$ to 30; confirmed on 1 October by an independent checker with its own code, which proved the three geometric facts for every d , among them that the extreme corner is unique, found the multipliers identical row for row, and wrote the hand lemma; no human reader). It is recorded as a fact about K_u , since Theorem 1 does not use it.

12 Cells with a coprime pair of extents

This section proves Theorem H, with no conjecture, on every cell (a, b, c) in which two of the three extents are coprime. The proof does not pass through Theorem U or K_u . It rests on a statement about the types of one sublattice, which holds in every solution of the type model and so needs neither the signs nor a hexagon of odd curl. The cubic cells are not covered, and the last subsection but one says what is different on them.

12.1 The parametrisation

The type model on the cell (a, b, c) is an assignment of types that satisfies (T2) on every hexagon, with coordinates modulo $(4a, 4b, 4c)$. Write X, Y and Z for the indicators of the three types. The project's working note proves, for every cell, that every real solution of the one-hot equations and of (T2) has the form

$$Y(v) = F_x(v) - F_z(v) + \sigma_v \alpha, \quad Z(v) = F_y(v) - F_x(v) + \sigma_v \beta, \quad (\text{C})$$

where a real weight is attached to every chain, $F_k(v)$ is the sum of the weights of the two chains of axis k through v , $\sigma_v = \pm 1$ on A and B, and α and β are constants. The chains of axis x have steps $(0, 2, 2)$ and $(0, 2, -2)$, those of axis y have $(2, 0, 2)$ and $(2, 0, -2)$, and those of axis z have $(2, 2, 0)$ and $(2, -2, 0)$. A chain of axis x lies in two neighbouring planes $x = \text{const}$ and closes after $2bc / \gcd(b, c)$ steps, so each such pair of planes holds $\gcd(b, c)$ chains of its family. When $\gcd(b, c) = 1$ it holds one, and $F_x(v) = \Phi(x_v)$ is a

function of the plane of v .

On a plane $z = z_0$ of a sublattice S the vertices are $v_0 + m(2, 2, 0) + n(2, -2, 0)$. The chain of axis z through such a vertex with step $(2, 2, 0)$ is labelled by n and the one with step $(2, -2, 0)$ by m , so on this plane

$$F_z = P(n) + Q(m), \quad x = x_0 + 2(m + n),$$

with P and Q of period $g_z = \gcd(a, b)$. The translations of the plane act transitively on its vertices, the step $(2, 2, 0)$ fixes the class of n and moves that of m by one, and the other step does the reverse. Every pair of classes therefore occurs at a vertex of the plane. The same holds on a plane $y = y_0$, with period $g_y = \gcd(a, c)$, and on a plane $x = x_0$.

12.2 Three lemmas on sums of functions

Lemma 8. Let $\varepsilon(i, j) = \alpha(i) + \beta(j)$ take the values 0 and 1 on a product $I \times J$. Then α or β is constant.

Proof. If $\alpha(i_1) \neq \alpha(i_2)$, the two differ by ± 1 , and for every j the values $\varepsilon(i_1, j)$ and $\varepsilon(i_2, j)$ are 0 and 1 in an order that does not depend on j . That determines $\beta(j)$. $\square$

Lemma 9. Let $\varepsilon(m, n) = \gamma(m + n) + \alpha(n) + \beta(m)$ take the values 0 and 1 on $\mathbb{Z}^2$, with α, β and γ real and periodic, and suppose that the jumps $\gamma(k + 1) - \gamma(k)$ are integers. Then either γ is constant and ε is a function of m alone or of n alone, or α and β are constant and ε is a function of $m + n$.

Proof. Let p, q and c be the jumps of α, β and γ . Since $\varepsilon(m + 1, n) - \varepsilon(m, n) = q(m) + c(m + n)$, the function q takes integer values, and so does p . Also $\varepsilon(m, n + 1) - \varepsilon(m + 1, n) = p(n) - q(m)$ lies in $\{-1, 0, 1\}$. The jumps of a periodic function add to zero over a period. If p is not zero it therefore has a value at least 1 and a value at most -1 , and then $0 \leq q(m) \leq 0$ for every m . So α or β is constant. If β is constant, ε is the sum of a function of $k = m + n$ and a function of n , and the pair (k, n) runs over $\mathbb{Z}^2$, so Lemma 8 makes γ or α constant. $\square$

The hypothesis on the jumps cannot be dropped. On $\mathbb{Z}_3 \times \mathbb{Z}_3$ take $\alpha = \beta = (\frac{2}{3}, \frac{1}{3}, 0)$ and $\gamma = (-\frac{1}{3}, 0, -\frac{2}{3})$. The sum $\alpha(i) + \beta(j) + \gamma(i + j)$ is 1 at the three points $(0, 0)$, $(0, 1)$ and $(1, 0)$ and 0 at the other six, and it is a function of none of i, j and $i + j$. On $\mathbb{Z}_2 \times \mathbb{Z}_2$

every function with values 0 and 1 has this form. The project's report of 18 September 2026 stated the lemma without the hypothesis, and the proof of Theorem R1 used it in that form. The next lemma supplies the hypothesis where Theorem R1 and this section need it, so Theorem R1 stands.

Lemma 10. Let $\gcd(b, c) = 1$, let S be a sublattice, and let Φ_S be the restriction of Φ to the planes $\$x = \$ \text{const}$ of S . In a solution of the type model the jumps $\Phi_S(x+2) - \Phi_S(x)$ are integers.

Proof. On a plane $z = z_0$ of S , (C) gives $Y = \gamma(m+n) - P(n) - Q(m)$ up to a constant, with $\gamma(k) = \Phi_S(x_0 + 2k)$. Let c and q be the jumps of γ and of Q , so that $c(m+n) - q(m)$ lies in $\{-1, 0, 1\}$ for all m and n . Suppose some value of c is not an integer, and let θ , with $0 < \theta < 1$, be its fractional part. Every value of q is then congruent to θ modulo 1, and so is every value of c . Each of the two functions has sum zero over a period and no value zero, so each has a value at least θ and a value at most $\theta - 1$. Since $\max c - \min q \leq 1$ and $\min c - \max q \geq -1$, both take the values $\theta - 1$ and θ and no others. If j of the g_z values of q in a period equal θ , then $j\theta + (g_z - j)(\theta - 1) = 0$, so $g_z\theta$ is an integer. On a plane $y = y_0$ of S the same argument, applied to Z , shows that $g_y\theta$ is an integer. The greatest common divisor of g_y and g_z is $\gcd(a, b, c)$, which is 1, so θ is an integer. $\square$

12.3 The theorem on one sublattice

Theorem 12. Let $\gcd(b, c) = 1$. In every solution of the type model on the cell (a, b, c) , and in particular in every frozen state, each sublattice S satisfies one of the following.

1. S has no vertex of type x , and every plane $\$x = \$ \text{const}$ of S is entirely of type y or entirely of type z .
2. S has no vertex of type y .
3. S has no vertex of type z .

So no sublattice carries all three types. By the permutations of the axes the same holds on every cell in which some pair of extents is coprime.

Proof. Absorb the constants of (C) on S into the heights, so that on S we have $Y = \Phi_S(x) - F_z$ and $Z = F_y - \Phi_S(x)$.

Suppose first that Φ_S is not constant. On a plane $z = z_0$ of S , Lemma 9 applies to Y , by Lemma 10, and its function γ is not constant, so P and Q are constant on that plane. Hence $Y(v) = \Phi_S(x_v) - \kappa(z_v)$ on S . Every pair (x, z) of coordinates of the parity of S belongs to a vertex of S , so Lemma 8 applies, and κ is constant. In the same way $Z(v) = \kappa' - \Phi_S(x_v)$ with κ' constant. The two values of Φ_S are κ and $\kappa + 1$ by the first formula and $\kappa' - 1$ and κ' by the second. So $\kappa' = \kappa + 1$ and $Y + Z = 1$ on S , which is the first case.

Suppose now that Φ_S is constant. On a plane $z = z_0$ of S the indicator Y is a constant minus $P(n) + Q(m)$, and by Lemma 8 it is a function of n alone or of m alone. Every vertex u of type y of S therefore lies on a line $\{u + t(2, 2\epsilon, 0)\}$, $\epsilon = \pm 1$, all of whose vertices are of type y . In the same way every vertex r of type z of S lies on a line $\{r + s(2, 0, 2\epsilon')\}$ all of type z . Two such lines have a common vertex. Write $u - r = (d_x, d_y, d_z)$, whose entries are even and add to a multiple of 4. A common vertex needs $t \equiv -\epsilon d_y/2$ modulo $2b$, $s \equiv \epsilon' d_z/2$ modulo $2c$, and $t - s \equiv -d_x/2$ modulo $2a$. The first two fix $t - s$ up to multiples of $2 \gcd(b, c) = 2$, and its parity is that of $(d_x + d_y + d_z)/2$, which is even. So $t - s$ can be given the value required. A common vertex would have two types, so S lacks type y or type z . $\square$

The hypothesis $\gcd(b, c) = 1$ is used to have one chain of axis x in each pair of planes. The last step of Lemma 10 and the meeting of the two lines need only $\gcd(a, b, c) = 1$.

Theorem 13. Let $\gcd(a, b) = \gcd(a, c) = 1$. In every solution of the type model each sublattice S has one of three forms.

1. S has types x and y only, and its vertices of type x are those on a set of planes $\$z = \$ \text{const.}$
2. S has types x and z only, and its vertices of type x are those on a set of planes $\$y = \$ \text{const.}$
3. S has no vertex of type x , and on each plane $\$x = \$ \text{const}$ of S the type is constant along the chains of axis x of one of the two families.

Proof. Now F_y is a function of y and F_z a function of z . By (C), $X = 1 + F_z - F_y$ up to a constant on S , a sum of a function of z and a function of y on the set of all pairs (y, z) of the parity of S . By Lemma 8 it is a function of z alone or of y alone. Suppose it is a function of z alone and is not zero. Then F_y is constant on S , some plane $z = z_1$ of S is all of type x , and $Z = F_y - F_x + \sigma\beta$ is, on each plane $x = x_0$ of S , a sum of a function of the chain

of one family and a function of the chain of the other. By Lemma 8 it is constant along the chains of one family. The line $z = z_1$ of that plane has its vertices 4 apart in y , and the chains of each family are the classes of $y - z$ or of $y + z$ modulo $4 \gcd(b, c)$, which divides $4b$, so the line meets every chain of both families. On the line $Z = 0$. Hence $Z = 0$ on the plane and on S , which is the first form. The second form arises in the same way, and when $X = 0$ on S the same use of Lemma 8 gives the third. $\square$

12.4 Theorem H

Theorem 14. On a cell in which some pair of extents is coprime, every odd frozen state lacks one of the three types. Theorem H therefore holds on the cell. If $\gcd(a, b) = \gcd(a, c) = 1$ the type that is absent is x .

Proof. By Lemma 1 an odd state has a corner O . Its two rods are methylenes of O , so they are vertices of the sublattice S of O , and they have two different types j and k . By Theorem 12, S has no vertex of the third type i . If the other sublattice had one, Theorem M' would give even curl. So no vertex has type i , and the axis i is empty.

Let $\gcd(a, b) = \gcd(a, c) = 1$. A corner of S cannot have a rod of type x . In the first form of Theorem 13 the rod of type x and the other rod, which is of type y and lies on the line through the hole along y , have the same coordinate z , and the plane of that coordinate is all of type x . The second form is excluded in the same way. So the rods of a corner have types y and z , and its sublattice has the third form. This gives a second proof of the theorem on these cells, which does not use Theorem M' . Let (O, σ) be the odd hexagon at the corner O , and $O' = O + \sigma$, a hole of the other sublattice S' . By (T2) none of the vertices $b_k = O' - 2\sigma_k e_k$ has type k . If S' had the first form, b_y would be of type x , the plane $z = z_{O'}$ of S' would be all of type x , and b_x , which lies in that plane, would be of type x . The second form is excluded by b_z and b_x . So S' has the third form and no vertex has type x . $\square$

With Lemma A and Theorem P, every frozen zero-flux state of a cell with a coprime pair of extents, all of them at least two, is invariant under a nonzero translation. The cells $(L, L, L + 1)$ are of this kind for every $L \geq 2$, with two coprime pairs. The thin cells $(1, b, c)$ have a coprime pair and are excluded for the reason given in the first section: on them the conclusion fails.

Theorem U holds on these cells as well. The unoriented cage has bridgeheads of two types on one sublattice, which by Theorem 12 has no vertex of the type of the rods, and Theorem M' gives even curl. Theorem 10 below is the case in which the coprime pair is transverse to the rods. Theorem 11 is not contained in this section, since it holds on cells such as $(15, 6, 10)$ that have no coprime pair.

The first proof of Theorem 14 uses K_u nowhere and Theorem U nowhere. It does use Theorem M' , whose proof rests on lemmas about four hexagons with DRAT certificates. Those lemmas are statements about finitely many vertices of the lattice and hold on every torus. The second proof uses no certificate.

12.5 The fractions of a type on the three lines through a vertex

Theorem 5 has a sharper form, which holds on every cell. For a vertex v let $h_k(v)$ be the number of vertices of type k on the line $\{v + 4ne_k\}$ and ℓ_k the number of vertices of that line.

Theorem 15. In a solution of the type model there are, on each sublattice, functions P , Q and R of one coordinate with

$$\frac{h_y}{\ell_y} = P(x) - R(z), \quad \frac{h_z}{\ell_z} = Q(y) - P(x), \quad \frac{h_x}{\ell_x} = 1 + R(z) - Q(y),$$

so that the three fractions at any vertex add to one.

Proof. Average (C) along the line through v in the direction y . The chains of axis x are labelled by $y - z$ or $y + z$ modulo $4 \gcd(b, c)$, which divides $4b$, so along the line the chain of each family runs equally often over all the chains of its pair of planes, and the average of F_x is a function $P(x)$. The average of F_z is a function $R(z)$ for the same reason. Along the line in the direction z the average of F_x is the same $P(x)$ and that of F_y is a function $Q(y)$. Along the line in the direction x the average of F_x is not a function of one coordinate, but it cancels from $X = 1 - Y - Z$. The constants of (C) are absorbed on each sublattice. $\square$

By Lemma 1 a frozen state has even curl exactly when every fraction is 0 or 1.

12.6 The cells with no coprime pair

The other cells are those in which every two extents have a common factor. The cubic cells of side at least two are among them, and so are $(2, 2, 4)$, $(4, 4, 6)$ and $(6, 10, 15)$. On such a cell every pair of planes holds at least two chains of each family and no height is a function of one coordinate. Theorem 12 is false there. Frozen states with an odd hexagon and all three types present exist on $(2, 2, 2)$, $(2, 2, 4)$, $(2, 4, 6)$, $(3, 3, 3)$ and $(4, 4, 6)$ (solver, with the sign field). Of the 1,357 odd states of the cubic cells of side three and four that the project has collected, 356 have two types on each sublattice and an empty axis, and 1,001 have all three types on both sublattices. Each of the 1,001 has an axis that is one-way and not empty. On these cells Theorem H is therefore a statement about the signs of the moments, and no statement about the types alone proves it. The next section gives the linear structure of the signed moments, which holds on every cell. The section after it proves Theorem H on the first family of these cells, those with two chains in every pair of planes. On the rest the reduction through Theorem U and K_u is the route that is known.

12.7 Checks

The proofs of this section were written by the main session of the project and read by a separate agent as checker, which found no gap. The checker wrote its own programs. It confirmed the counterexample to the lemma on three families in exact arithmetic, and enumerated the meeting of the lines of Theorem 12 on twelve cells. In the type model the statement that one sublattice has a vertex of each type is unsatisfiable on $(2, 3, 3)$, $(4, 3, 3)$, $(5, 3, 3)$, $(3, 4, 4)$, $(5, 2, 4)$, $(5, 4, 4)$, $(6, 2, 3)$, $(6, 3, 4)$, $(6, 2, 9)$ and $(7, 3, 6)$, by two encodings, and satisfiable on $(3, 3, 3)$, $(4, 4, 6)$ and $(2, 2, 4)$. On $(10, 4, 5)$ the solver gave no verdict. On 258 sampled solutions of seven cells with a coprime pair, the case of Theorem 12, the forms of Theorem 13 and the integrality of Lemma 10 had no exception, and Theorem 15 held exactly on 495 sampled solutions of ten cells that are not cubic and on the 1,357 states of the cubic cells. These computations are tests of the proofs and are not steps of them.

13 The linear structure of the signed moments

The parametrisation (C) describes the types of a frozen state by weights on the chains. The signed moments have a description of the same kind in any state whose holes are all oriented, which by Theorem U is every odd state on the cells where that theorem is known. On the cells with no coprime pair Theorem H is a statement about signs, so a proof there has to use this structure. The next section uses it for the cells with two chains in every pair of planes.

13.1 Notation

A chain of the family $f = \{i, j\}$ alternates bonds of the directions d_i and d_j , and its step is $u_f = d_i - d_j$. The six steps are

$$\begin{aligned} u_{01} &= (0, 2, 2), & u_{02} &= (2, 0, 2), & u_{03} &= (2, 2, 0), \\ u_{23} &= (0, 2, -2), & u_{13} &= (2, 0, -2), & u_{12} &= (2, -2, 0), \end{aligned}$$

two for each axis, and the step of a chain has no component along the axis of the chain. Two distinct directions have $d_r \cdot d_s = -1$, so $d_r \cdot u_f$ is 4 for $r = i$, -4 for $r = j$ and 0 for the two directions that the chain does not use.

Let Λ be the lattice of translations of A , generated by the steps, and G its quotient by the periods of the cell (a, b, c) , a group of order $4abc$. A character of G extends to $\chi(p) = \xi^{p_1} \eta^{p_2} \zeta^{p_3}$ with $\xi^{4a} = \eta^{4b} = \zeta^{4c} = 1$. Fix one extension for each character, and put $\lambda_r = \chi(d_r)$. The functions $\chi \mathbf{1}_A$ and $\chi \mathbf{1}_B$ form a basis of the complex functions on the vertices. Every linear system below has integer coefficients and is invariant under Λ . It therefore acts on each plane spanned by $\chi \mathbf{1}_A$ and $\chi \mathbf{1}_B$ separately, and its real solutions are known once its complex solutions are known for every character.

Say that the family f is supported at χ if $\chi(u_f) = 1$, that is, if $\lambda_i = \lambda_j$. The indicator of a chain of the family f has a component at χ only if f is supported there, and the component is then a multiple of $(1, 1/\lambda_i)$ in the basis above.

Lemma 11. The families supported at a character are the pairs inside the blocks of the partition of $\{0, 1, 2, 3\}$ by the value of λ . There are five cases: no family; one family; the two families of one axis; three families, one of each axis, when three of the λ_r are equal;

and all six, which happens for the trivial character only.

Proof. The complementary pairs $\{01, 23\}$, $\{02, 13\}$ and $\{03, 12\}$ are the pairs of families of the axes x , y and z . If all four λ_r are equal, χ is trivial on every step and hence on Λ . $\square$

13.2 One cage law

Lemma 12. The indicator of a chain of an axis other than k satisfies law P^k at every hole, and so do $\mathbf{1}_A$ and $\mathbf{1}_B$.

Proof. Take $k = x$ and the A-hole $O = a + 2e_x$. Its bridgeheads are $a + d_0$ and $a + d_1$ on the lower side and $a + 4e_x + d_2$ and $a + 4e_x + d_3$ on the upper, so the law reads

$$K(a + 4e_x) - K(a) = K(a + 4e_x + d_2) + K(a + 4e_x + d_3) - K(a + d_0) - K(a + d_1).$$

Let C be a chain of the family $\{0, 2\}$. A vertex b of B lies on C exactly when $b - d_0$ does, and exactly when $b - d_2$ does. So $K(a + d_0) = K(a)$ and $K(a + 4e_x + d_2) = K(a + 4e_x)$ for $K = \mathbf{1}_C$. The two remaining terms are the values of K at $a + d_1$ and $a + 4e_x + d_3$, which lie on C exactly when $a + (0, -2, -2)$ and $a + (4, -2, 2)$ do. These two points differ by $2u_{02}$, so the terms are equal. The three other families of the axes y and z are the same computation. For the constants both sides vanish. The inversion carries the law at a B-hole to the law at an A-hole and each family to itself, and the permutations of the coordinates permute the laws. $\square$

Theorem 16. On every cell the real solutions of law P^k , taken at every hole, are the span of the indicators of the chains of the two axes other than k and of $\mathbf{1}_A$ and $\mathbf{1}_B$. For $k = x$ the dimension is

$$4b \gcd(a, c) + 4c \gcd(a, b) - 4 \gcd(a, b, c) + 2.$$

Proof. Take $k = x$. On the plane of a character the law at the A-holes and at the B-holes is the matrix

$$M = \begin{pmatrix} \Delta & -\Theta \\ -\Theta' & \Delta \end{pmatrix}, \quad \Delta = \xi^2 - \xi^{-2}, \quad \Theta = \xi p - \xi^{-1} q, \quad \Theta' = \xi q - \xi^{-1} p,$$

with $p = \eta/\zeta + \zeta/\eta$ and $q = \eta\zeta + 1/(\eta\zeta)$. Write $X_\xi = \xi^2 + \xi^{-2}$ and likewise for η and ζ . Then $pq = X_\eta + X_\zeta$ and $p^2 + q^2 = 4 + X_\eta X_\zeta$, so

$$\det M = \Delta^2 - \Theta\Theta' = (X_\xi - X_\eta)(X_\xi - X_\zeta),$$

and $X_\xi - X_\zeta = \xi^{-2}(\chi(u_{02}) - 1)(\chi(u_{13}) - 1)$, $X_\xi - X_\eta = \xi^{-2}(\chi(u_{03}) - 1)(\chi(u_{12}) - 1)$. The determinant vanishes exactly when some family of the axes y and z is supported. If $\Delta = 0$ then $\xi^2 = \varsigma = \pm 1$, and $\xi\eta\zeta\Theta = -(\eta^2 - \varsigma)(\zeta^2 - \varsigma)$ and $\xi\eta\zeta\Theta' = \varsigma(\eta^2 - \varsigma)(\zeta^2 - \varsigma)$. So $M = 0$ exactly when $\xi^2 = \eta^2 = \pm 1$ or $\xi^2 = \zeta^2 = \pm 1$, which says that both families of z , or both families of y , are supported.

By Lemma 12 the component of every supported chain of y or z lies in the kernel of M . If no such family is supported the kernel is zero. If some are supported but not both families of one axis, the kernel is a line and any supported chain spans it. If both families of y are supported, the kernel is the whole plane, and the two components $(1, 1/\lambda_0)$ and $(1, 1/\lambda_1)$ are independent unless $\lambda_0 = \lambda_1$. In that case all four λ_r are equal, the character is trivial, and the two constants span the plane. If both families of z are supported the same holds with $\lambda_0 = \lambda_3$ and $\lambda_1 = \lambda_2$. So the kernel is the span at every character.

The dimension is the number of characters with $\det M = 0$ plus the number with $M = 0$. The characters at which a given family of y is supported number $2b \gcd(a, c)$, which is the number of chains of the family, and $2c \gcd(a, b)$ for a family of z . Both families of y are supported at $2b$ characters and both of z at $2c$. A family of y and a family of z are supported together at $\gcd(a, b, c)$ characters, those trivial on a plane of Λ normal to one of the d_r . Any three conditions force the fourth, and all four hold at the trivial character only. Counting by inclusion and exclusion gives the formula. $\square$

13.3 The three laws and the bond rule

By Lemma 3 the component n_k of the moments obeys law P^k at every oriented hole. The moments also obey the bond rule, $d \cdot n(v) = d \cdot n(v + d)$ on every bond, which is linear in n .

Theorem 17. On every cell the real vector fields n on the vertices whose components obey the three laws P^x , P^y and P^z at every hole, and which obey the bond rule on every

bond, are exactly the fields

$$n = \frac{1}{2} \sum_C g(C) u_C \mathbf{1}_C,$$

the sum being over all chains, with real weights g and u_C the step of the chain. The dimension of the space is $4(a \gcd(b, c) + b \gcd(a, c) + c \gcd(a, b)) - 4 \gcd(a, b, c) + 1$.

So each chain carries a current along itself, and the moments of a state with every hole oriented are a sum of such currents.

Proof. The field $\frac{1}{2}u_f \mathbf{1}_C$ obeys the bond rule. On a bond of a direction that C does not use both sides vanish, and a bond of the direction d_i or d_j with one end on C has both ends on C , where both sides equal $\frac{1}{2}d \cdot u_f$. Its component along the axis of C is zero, and each of the two others is a multiple of $\mathbf{1}_C$, which obeys the law of that component by Lemma 12.

Conversely, fix a character and write n_A and n_B for the two amplitudes, vectors of three components. The bond rule reads $d_r \cdot n_A = \lambda_r d_r \cdot n_B$ for $r = 0, \dots, 3$, and the laws say that each pair of components lies in the kernel of the matrix of Theorem 16 for its axis. Take the cases of Lemma 11 in turn. If no family is supported, every determinant is non-zero and $n = 0$. If one family $f = \{i, j\}$ of axis m is supported, then $n_m = 0$, the laws of the two other components give $n_B = n_A / \lambda_i$, and the bond rule becomes $(1 - \lambda_r / \lambda_i) d_r \cdot n_A = 0$. For the two directions r outside f the factor is not zero, so n_A is orthogonal to both and is a multiple of u_f . If both families of the axis m are supported, with values $\lambda = \mu$ on one and $\lambda = \nu$ on the other, then $n_m = 0$, the laws of the two other components say nothing, and the bond rule leaves two independent equations on four unknowns. The solutions form a plane, and the currents of the two families are two independent vectors of it. If three families are supported, with $\lambda_r = \mu$ for $r \neq \ell$, the laws give $n_B = n_A / \mu$, and the bond rule on the direction d_ℓ gives $n_A \cdot d_\ell = 0$, a plane spanned by the steps of the three families. At the trivial character the laws say nothing and the bond rule gives $n_A = \lambda n_B$, three dimensions, spanned by the six steps. In every case the solutions are the currents. The dimension is the number of chains less the number of relations. There is one relation at each non-trivial character at which three families are supported, $4(\gcd(a, b, c) - 1)$ in all, and there are three at the trivial character. $\square$

In components, with $G_f(v)$ the weight of the chain of the family f through v ,

$$n_x = G_{02} + G_{13} + G_{03} + G_{12},$$

$$n_y = G_{01} + G_{23} + G_{03} - G_{12},$$

$$n_z = G_{01} - G_{23} + G_{02} - G_{13}.$$

The two families of an axis enter the two transverse components with different signs, one with $(+, +)$ and the other with $(+, -)$. In (C) both families of an axis enter the two transverse types with $(+, -)$. No constant on one sublattice alone occurs, as the terms with σ_v do in (C). A field that is constant on all the vertices is a sum of currents: e_x is the sum over the chains of the two families of axis y with every weight $\frac{1}{2}$. Neither system implies the other: the three laws never imply the bond rule, and the bond rule implies the laws on the cell $(1, 1, 1)$ only.

Theorem 18. In a frozen state with every hole oriented, each of the six indicators of a direction is a sum of weights over the chains through the vertex, and a constant on each sublattice. Each chain carries two weights. Each weight belongs to a pair of directions transverse to the axis of the chain whose difference is half the step of the chain, and enters the indicator of one with the sign $+$ and that of the other with $-$.

Proof. Let $f(C)$ be the weights of (C) and $g(C)$ those of Theorem 17, which applies by Lemma 3. The indicator of the direction $\pm e_k$ is $\frac{1}{2}(K_k \pm n_k)$. For a chain of the family $\{0, 1\}$ the directions transverse to its axis fall into the pairs $\{+e_y, -e_z\}$ and $\{+e_z, -e_y\}$, in each of which the difference is $\pm(0, 1, 1)$. Giving the first pair the weight $\frac{1}{2}(f + g)$ and the second $\frac{1}{2}(g - f)$ reproduces the contribution of the chain to the types, which is $f(C)$ to Y and $-f(C)$ to Z , and its contribution $\frac{1}{2}g(C)u_C$ to n . The other families are the same. $\square$

The theorem is linear and says nothing about integers. It does not contain the relation $|n_k| = K_k$ between the two sets of weights.

13.4 The signed fractions on the lines through a vertex

Let $\sigma_k(v)$ be the sum of n_k over the line $\{v + 4ne_k\}$ divided by the number of its vertices.

Theorem 19. In a frozen state with every hole oriented there are six functions of one

coordinate, F_{01} and F_{23} of x , U_{02} and U_{13} of y , W_{03} and W_{12} of z , with

$$\begin{aligned}\sigma_x &= [U_{02} + U_{13}](y) + [W_{03} + W_{12}](z), \\ \sigma_y &= [F_{01} + F_{23}](x) + [W_{03} - W_{12}](z), \\ \sigma_z &= [F_{01} - F_{23}](x) + [U_{02} - U_{13}](y)\end{aligned}$$

at the vertices of A, and with the same six functions at the vertices of B, evaluated at $x - 1$, $x + 1$, $y - 1$, $y + 1$, $z - 1$ and $z + 1$ in the order in which they are named.

Proof. Average the components of Theorem 17 along the line, as in the proof of Theorem 15. The average of G_f is the mean weight of the chains of the family f in the pair of planes of v , and the pair of planes of a family lies above a vertex of A and below a vertex of B, or the reverse, according to the sign of the component of d_i along the axis. $\square$

The separability of each σ_k is the second statement of Theorem 5. What is added is that the three share their profiles, and that the fractions on B are those on A up to constants, which has no counterpart for the types. The three signed fractions at a vertex do not add to a constant. Their sum on A is $2F_{01}(x) + 2U_{02}(y) + 2W_{03}(z)$, and it is constant in 86 of the 1,357 odd states of the cubic cells of side three and four.

13.5 States with charged holes, and one sublattice

By Lemma 3 the component n_k fails law P^k at a charged hole by the charge, on every axis k and not only on the axis of the rods of the hole. In 62 frozen states of even curl sampled by solver on four cells, 45 of them with charged holes, the types obeyed the laws at every hole, the signed components failed by exactly the charge, and n_k lay in the span of Theorem 16 exactly when no hole was charged. So Theorems 17 to 19 are statements about oriented states and fail without that hypothesis.

Theorem 12 has no counterpart for directions. On the cells $(4, 3, 3)$, $(6, 2, 3)$ and $(5, 2, 4)$ one sublattice of a frozen state can carry any pattern of absent, one-way and two-way axes on two of the three axes (solver, with a satisfiable control for each query), so the signs add nothing on one sublattice to what Theorem 12 says about the types. A statement of the kind of K_u , that some direction is absent, has to couple the two sublattices, and Theorem 19 is one place where they are coupled.

13.6 Checks

The theorems of this section were found and proved by an agent of the project. The main session redid the algebra of the determinant and of the cases of Theorem 17 and ran the agent's programs again. Exact ranks modulo a prime agree with Theorem 16 on twenty cells and with Theorem 17 on twenty-two, among them cells of extent one and two, cells with a coprime pair and cells with none, and for Theorem 17 an independent count over the characters gives the same dimensions. Every one of the 1,357 odd states of the cubic cells of side three and four, and 48 sampled odd states of four cells that are not cubic, satisfies the three laws and the bond rule, and Theorem 19 holds exactly on all of them. A separate agent then read the section as checker, with its own programs, and found no gap. It made three corrections of wording, which are in the text above: on constants, on the sum of the three signed fractions, and on one case of the proof of Theorem 16. Its exact ranks agree with Theorems 16 and 17 on sixteen cells, and Theorem 19 holds exactly on 837 odd states of nine cells and fails on 773 of them when the six shifts are reversed.

14 Cells with two chains in every pair of planes

This section concerns the cells $(2a', 2b', 2c')$ in which a', b' and c' are pairwise coprime, such as $(2, 2, 2)$, $(2, 4, 6)$, $(4, 6, 10)$ and $(6, 10, 14)$. Every two extents of such a cell have greatest common divisor 2, so no pair is coprime and the section on cells with a coprime pair does not apply. Every pair of neighbouring planes holds exactly two chains of each family, for each of the three axes. The section proves statement (V) on these cells, and with it Theorem U, and then Theorem H. Neither proof uses K_u . They are the first proofs of (V) and of Theorem H on cells with no coprime pair, and the first proof of Theorem H in which an axis that is one-way and not empty is produced by an argument and not by a certificate.

14.1 Signs, classes and chains

At a vertex $v = (x, y, z)$ of A put $\rho_x = (-1)^{x/2}$, $\rho_y = (-1)^{y/2}$, $\rho_z = (-1)^{z/2}$ and $\pi = (-1)^{(x+y+z)/4}$. At a vertex of B put $\rho_x = (-1)^{(x-1)/2}$, likewise ρ_y and ρ_z , and $\pi = (-1)^{(x+y+z-3)/4}$. The periods of the cell are multiples of 8, so these are defined on the cell, and $\rho_x \rho_y \rho_z = 1$ at every vertex. The four triples of signs divide each sublattice

into four classes. A vertex and its translate by $4e_k$ lie in the same class and have opposite π .

The two chains of a family in a pair of planes are the two classes modulo 8 of $y - z$ or of $y + z$, and likewise for the other axes. They are told apart by a sign that is constant along the chain. At the vertices of A it is $\pi\rho_z$ for the family $\{0, 1\}$, $\pi\rho_x$ for $\{0, 2\}$, $\pi\rho_y$ for $\{0, 3\}$, and π for $\{2, 3\}$, $\{1, 3\}$ and $\{1, 2\}$. For instance $\pi\rho_z = (-1)^{(x+y-z)/4}$, and $x + y - z$ is unchanged by the step $(0, 2, 2)$ and changes by 4 from one chain of the pair of planes to the other. Write the weight of the chain of the family f whose vertices of A lie on the plane n and whose sign is s as $m_f(n) + s e_f(n)$.

Theorem 20. For a plane n of A and a sign σ put

$$E_x^A(n, \sigma) = \sigma e_{01}(n) + e_{23}(n),$$

$$E_y^A(n, \sigma) = \sigma e_{02}(n) + e_{13}(n),$$

$$E_z^A(n, \sigma) = \sigma e_{03}(n) + e_{12}(n),$$

and for a plane n' of B put $E_x^B(n', \sigma) = \sigma e_{01}(n' - 1) - e_{23}(n' + 1)$, and likewise for y and z . Let $\Phi_x^A(n) = m_{01}(n) + m_{23}(n)$ and $\Phi_x^B(n') = m_{01}(n' - 1) + m_{23}(n' + 1)$, and likewise for y and z . Then on each sublattice S

$$F_x = \Phi_x(x) + \pi E_x(x, \rho_z),$$

$$F_y = \Phi_y(y) + \pi E_y(y, \rho_x),$$

$$F_z = \Phi_z(z) + \pi E_z(z, \rho_y),$$

and every real solution of the one-hot equations and of (T2) is

$$Y = a_y + \pi b_y, \quad Z = a_z + \pi b_z, \quad X = a_x + \pi b_x,$$

with $a_y = \Phi_x(x) - \Phi_z(z) + \sigma_S \alpha$, $a_z = \Phi_y(y) - \Phi_x(x) + \sigma_S \beta$, $a_x = 1 - a_y - a_z$, and $b_y = E_x - E_z$, $b_z = E_y - E_x$, $b_x = E_z - E_y$, each E_k being taken at the plane and the sign of the vertex.

Proof. At a vertex of A the chain of $\{0, 1\}$ through v has its vertices of A on the plane x and sign $\pi\rho_z$, and the chain of $\{2, 3\}$ has the same plane and sign π , which gives F_x . The

two other axes are the same. At a vertex w of B the chain of $\{0, 1\}$ contains $w - d_0$, on the plane $x - 1$ of A, and its sign there is $(-1)^{(x+y-z-1)/4}$, which is $\pi\rho_z$ at w because the two exponents differ by the even number $z - 1$. The chain of $\{2, 3\}$ contains $w - d_2$, on the plane $x + 1$, with sign $(-1)^{(x+y+z+1)/4} = -\pi$. The other four families are the same computation. Substituting in (C) gives the rest. $\square$

Since $\rho_y = \rho_x\rho_z$, the function b_y depends on x and z only, and likewise b_z on x and y and b_x on y and z . Four consequences follow for a solution of the type model.

Lines. Along a line $\{v + 4ne_k\}$ the two transverse coordinates are fixed and π alternates. The indicator K_k takes the values 0 and 1, so the pair (a_k, b_k) is $(0, 0)$, $(1, 0)$ or $(\frac{1}{2}, \pm\frac{1}{2})$. The type k is absent from the line, fills it, or occupies every second vertex. The number a_k is the fraction of Theorem 15.

Profiles. By Theorem 15 there are functions p, q and r of one coordinate on S with $2a_y = p(x) - r(z)$, $2a_z = q(y) - p(x)$ and $2a_x = 2 + r(z) - q(y)$. Every pair of values of two coordinates occurs at a vertex of S , so the differences are integers, and the profiles can be taken to be integers. Since the fractions are not negative,

$$\max r \leq \min p, \quad \max p \leq \min q, \quad \max q \leq \min r + 2.$$

The three gaps in these inequalities and the three spreads $\max - \min$ of r, p and q are six integers, none negative, that add to 2. In particular the three spreads add to at most 2.

Parity. A line has fraction $\frac{1}{2}$ exactly when $|b_k| = \frac{1}{2}$. So $|E_x - E_z| = \frac{1}{2}$ at a vertex if $p(x) - r(z)$ is odd and $E_x = E_z$ if it is even, and likewise $|E_y - E_x|$ with $q - p$ and $|E_z - E_y|$ with $r - q$.

The two sublattices. Put $\Delta_k(n) = E_k(n, +) - E_k(n, -)$ and $\Sigma_k(n) = E_k(n, +) + E_k(n, -)$. By the definitions

$$\Delta_k^B(n') = \Delta_k^A(n' - 1), \quad \Sigma_k^B(n') = -\Sigma_k^A(n' + 1).$$

The functions E of B are those of A, moved by one plane. The profiles p, q, r of B are not tied to those of A.

14.2 The unoriented cage

Take the cage in the position of Theorem 11, with the hole $O = (2, 0, 0)$, the rods $v = (0, 0, 0)$ and $v' = (4, 0, 0)$, the lower bridgeheads $\beta = (1, 1, 1)$ and $\beta' = (1, -1, -1)$ and the upper bridgeheads $\gamma = (3, 1, -1)$ and $\gamma' = (3, -1, 1)$.

Theorem 21. On a cell $(2a', 2b', 2c')$ with a', b', c' pairwise coprime, let a solution of the type model have v and v' of type x , β and β' of type y , and γ and γ' not of type y . Then sublattice B has no vertex of type x , its plane $x = 1$ is entirely of type y , and its plane $x = 3$ is entirely of type z .

Proof. The profiles and the functions E without a superscript are those of A in the first two steps.

The rod line. The rods lie on one line in the direction x and have opposite π . Both are of type x , so the line is full, $a_x = 1$ on it, and $q(0) = r(0)$. The inequalities give $r(0) \leq \min p \leq \max p \leq q(0)$, so p is a constant c on A, $r \leq c$ with equality at $z = 0$, and $q \geq c$ with equality at $y = 0$. On A the fraction a_y is a function of z and a_z a function of y , and both vanish at o.

The lines through the origin. At the vertices of A with $z = 0$ the fraction a_y is zero, so $b_y = 0$, and there $\rho_z = 1$ and $\rho_y = \rho_x$. Hence $E_x(x, +) = E_z(0, \rho_x)$. So $E_x(n, +)$ has one value $\kappa = E_z(0, +)$ on the planes $n \equiv 0$ and one value $\lambda = E_z(0, -)$ on the planes $n \equiv 2$ modulo 4. At the vertices with $y = 0$, $a_z = 0$ gives $E_y(0, \rho_x) = E_x(x, \rho_x)$, so $E_x(n, -)$ has one value $\mu = E_y(0, -)$ on the planes $n \equiv 2$, and $E_y(0, +) = \kappa$.

The bridgeheads. The classes (ρ_x, ρ_y, ρ_z) of β, β', γ and γ' are $(+, +, +), (+, -, -), (-, +, -)$ and $(-, -, +)$, and π is $+, -, +$ and $+$. By Theorem 20, with the profiles and the functions of B,

$$\begin{aligned} Y(\beta) + Y(\beta') - Y(\gamma) - Y(\gamma') &= p^B(1) - p^B(3) \\ &\quad + \Delta_x^B(1) - \Sigma_x^B(3) + \Sigma_z^B(-1) - \Delta_z^B(1). \end{aligned}$$

The left side is 2. By the relation between the two sublattices the last four terms are $\Delta_x^A(0) + \Sigma_x^A(4) - \Sigma_z^A(0) - \Delta_z^A(0)$, and $\Sigma_z^A(0) + \Delta_z^A(0) = 2\kappa$. With the values just found for $E_x(0, +)$ and $E_x(4, +)$ this gives

$$p^B(1) - p^B(3) = 2 + E_x^A(0, -) - E_x^A(4, -). \quad (\star)$$

The spread of p^B is at most 2. If the two values of E_x^A in $(\star)$ are equal, the spread is 2, so the five other integers vanish: r^B is the constant $p^B(3)$ and q^B the constant $p^B(1) = r^B + 2$. Then $a_x = 0$ on B, the fraction a_y is 1 on the plane $x = 1$, and a_z is 1 on the plane $x = 3$, which is the theorem. It remains to prove that the two values are equal. At odd length of the rod axis Theorem 11 had a vertex P of the rod line for this step, and at even length there is none.

The case in which some plane is integral. Suppose some plane $z = z_0$ of A with $z_0 \equiv 2$ modulo 4 has $a_y = 0$ or 1. The vertices (n, y, z_0) with $n \equiv 0$ are in the class $(+, -, -)$, and $b_y = 0$ there, so $E_x(n, -) = E_z(z_0, -)$ for every $n \equiv 0$, and the two values are equal. The same holds if some plane $y = y_0$ with $y_0 \equiv 2$ has $a_z = 0$ or 1, by $b_z = 0$ at the vertices (n, y_0, z) .

The case in which none is. Suppose $a_y = \frac{1}{2}$ on every plane $z \equiv 2$ of A and $a_z = \frac{1}{2}$ on every plane $y \equiv 2$. In the class $(+, -, -)$ the fraction a_x is then zero, so $E_z(z, -) = E_y(y, +)$ for all $y, z \equiv 2$, and both are a constant θ . In the same class $|b_y| = \frac{1}{2}$, so $E_x(n, -) = \theta \pm \frac{1}{2}$ for $n \equiv 0$. The difference in $(\star)$ is therefore $-1, 0$ or 1 . It is not 1, since the spread of p^B is at most 2, and if it is 0 the proof is finished. Suppose then that $E_x(0, -) = \theta - \frac{1}{2}$ and $E_x(4, -) = \theta + \frac{1}{2}$, so that $p^B(1) - p^B(3) = 1$. Three further facts hold on A. In the class $(-, +, -)$, $E_z(z, +) = \mu + \frac{1}{2}\psi(z)$ with $\psi(z) = \pm 1$, for $z \equiv 2$. In the class $(-, -, +)$, $E_y(y, -) = \lambda + \frac{1}{2}\varphi(y)$ with $\varphi(y) = \pm 1$, for $y \equiv 2$. And for a plane $n \equiv 0$ in the direction z , the numbers $E_z(n, +) - \kappa$ and $E_z(n, -) - \lambda$ are b_y , up to sign, in the classes $(+, +, +)$ and $(-, -, +)$, so both are zero or both are $\pm \frac{1}{2}$, and likewise $E_y(n, +) - \kappa$ and $E_y(n, -) - \mu$.

From the relation between the sublattices and the values found,

$$E_x^B(1, \sigma) - E_x^B(5, \sigma) = \frac{\sigma}{2}, \quad E_x^B(3, \sigma) - E_x^B(-1, \sigma) = -\frac{1}{2},$$

for both signs σ . Take a plane z' of B and a class of B that contains vertices of it, with $\rho_x = 1$. The vertices of that class on the planes $x = 1$ and $x = 5$ have $b_y = E_x^B(1, \rho_z) - h$ and $E_x^B(5, \rho_z) - h$, where $h = E_z^B(z', \rho_y)$. These two numbers differ by $\frac{1}{2}$ and each is 0 or $\pm \frac{1}{2}$, so one of them is 0 and h is $E_x^B(1, \rho_z)$ or $E_x^B(5, \rho_z)$. For a class with $\rho_x = -1$ the planes $x = 3$ and $x = -1$ do the same. For $z' \equiv 1$ modulo 4 the two classes are $(+, +, +)$

and $(-, -, +)$, and this gives

$$E_z^B(z', +) - E_x^B(1, +) \in \{0, -\frac{1}{2}\}, \quad E_z^B(z', -) - E_x^B(-1, +) \in \{0, -\frac{1}{2}\}.$$

Since $E_x^B(1, +) - E_x^B(-1, +) = \kappa - \lambda$, the difference $\Delta_z^A(z' - 1) = \Delta_z^B(z')$ is $\kappa - \lambda$ or differs from it by $\frac{1}{2}$. By the third fact it differs from $\kappa - \lambda$ by 0 or ± 1 , so it equals it. For $z'' \equiv 3$ the classes are $(+, -, -)$ and $(-, +, -)$, and the same argument, with $E_x^B(1, -) + E_x^B(-1, -) = -\kappa - \lambda$, gives $\Sigma_z^A(z'' + 1) = \kappa + \lambda$. Every plane $n \equiv 0$ is $z' - 1$ for one z' and $z'' + 1$ for one z'' . So $E_z^A(n, +) = \kappa$ and $E_z^A(n, -) = \lambda$ for every $n \equiv 0$, and in the same way $E_y^A(n, +) = \kappa$ and $E_y^A(n, -) = \mu$.

Now let $z' \equiv 1$ and $n = z' + 1$. On the plane $x = 1$ of B the vertices with coordinate z' are in the class $(+, +, +)$ and those with $z' + 2$ in the class $(+, -, -)$, and the values found give

$$\begin{aligned} E_x^B(1, +) - E_z^B(z', +) &= \frac{1}{4}(1 + \psi(n)), \\ E_x^B(1, -) - E_z^B(z' + 2, -) &= \frac{1}{4}(\psi(n) - 1). \end{aligned}$$

Exactly one of these is $\pm \frac{1}{2}$ and the other is 0. By the rule of parity $p^B(1) - r^B(z')$ and $p^B(1) - r^B(z' + 2)$ have different parities, so r^B is not constant. In the same way, with φ in place of ψ , q^B is not constant, and $p^B(1) - p^B(3) = 1$. The three spreads of B are each at least 1 and add to at least 3. This contradicts the inequalities. $\square$

The proof uses the parametrisation (C) and the count of two chains in a pair of planes. It uses that the types are integers through the three values of a fraction and the rule of parity, and that they are not negative through the inequalities on the profiles. It uses no certificate and no other result of this note.

Other positions. The inversion $w \mapsto (1, 1, 1) - w$ exchanges the sublattices. The half-turn $(x, y, z) \mapsto (4 - x, -y, z)$ fixes the hole, preserves the sublattices, the bonds and the types, and exchanges the two lower bridgeheads with the two upper. The exchange of y and z carries the cell to $(2a', 2c', 2b')$ and exchanges the two types of the bridgeheads. A permutation of the coordinates carries rods along y or z to rods along x . Each maps the family to itself, so the theorem holds for every unoriented cage of every cell of the family.

Theorem 22. On a cell $(2a', 2b', 2c')$ with a', b', c' pairwise coprime, every hole of an odd frozen state is oriented.

Proof. A frozen state with an unoriented hole has an unoriented cage, and by Theorem 21 the sublattice of its bridgeheads has no vertex of the type of its rods. The other sublattice has one, so Theorem M' gives even curl. An odd state has therefore no unoriented straight hole, and its corners are oriented by Lemma 2. $\square$

Theorem H follows from Theorem 22 and Theorem 1 on every cell of the family, since Theorem 1 in its form of 30 September needs no K_u . (In its first form it did, and K_u is established on these cells: the first step covers the slab when the greatest common divisor of the two extents transverse to the chain is at most 2, which is the case for every axis here, and the windows of its certificates were recorded for extents of at least three along the chain and four across it.) The next subsection gives the proof of Theorem H on these cells that was found first, on 28 September, and which does not pass through Theorem 1; it is kept because its case analysis says more than Theorem 1 does about which axis is one-way.

On the cells of the family with an extent of 2, among them $(2, 2, 2)$ and $(2, 4, 6)$, Theorem M' is used at an extent of 2. There it has been argued from lemmas on the infinite lattice. It has also been tested by solver in the type model on $(2, 2, 2)$, $(2, 2, 6)$, $(2, 4, 6)$, $(4, 2, 6)$ and $(2, 6, 10)$, for every axis and both sublattices, with satisfiable controls.

The proof of Theorem 21 does not extend as it stands to a cell such as $(4, 4, 6)$, in which the chains of one axis number four in a pair of planes. There the type y on a line of A need not have period 8. Statement (V) is true on $(4, 4, 6)$ and $(8, 4, 6)$ by solver.

14.3 Theorem H on these cells

Take an odd frozen state on a cell of the family. If some type is absent its axis is empty, so suppose that all three types occur. If one sublattice lacked a type that the other has, Theorem M' would give even curl. So both sublattices carry all three types. By Theorem 22 every hole is oriented, so Theorems 17 to 19 apply.

For a sublattice S write R , P and Q for the spreads of the profiles r , p and q , and $g_{rp} = \min p - \max r$, $g_{pq} = \min q - \max p$ and $g_{qr} = \min r + 2 - \max q$ for the three gaps. The six integers are not negative and add to 2.

Lemma 13 (six cases). If S carries all three types, exactly two of the six integers are 1, and they are either two of the spreads or a spread and the gap between the two other profiles. Name the cases by the coordinate k whose profile is constant in the first kind, and

whose lines all alternate in the second.

case	equal to 1	fractions
(i) _x	R, Q	a_y depends on z , and a_z on y
(i) _y	R, P	a_z depends on x , and a_x on z
(i) _z	P, Q	a_x depends on y , and a_y on x
(ii) _x	P, g_{qr}	$a_x = \frac{1}{2}$; a_y and a_z depend on x
(ii) _y	Q, g_{rp}	$a_y = \frac{1}{2}$; a_z and a_x depend on y
(ii) _z	R, g_{pq}	$a_z = \frac{1}{2}$; a_x and a_y depend on z

In a case (i)_k the two fractions named take the values 0 and $\frac{1}{2}$ each, and a_k is one less their sum, so it depends on both transverse coordinates. In a case (ii)_k the two fractions named add to $\frac{1}{2}$ and each takes both values.

Proof. The greatest value of $p - r$ is $R + g_{rp} + P$, so type y occurs on S exactly when that sum is at least 1. In the same way type z needs $P + g_{pq} + Q \geq 1$ and type x needs $Q + g_{qr} + R \geq 1$. The six integers lie round a circle in the order $R, g_{rp}, P, g_{pq}, Q, g_{qr}$, and the three conditions are on the three arcs of three consecutive terms that begin and end at a spread. A spread lies on two arcs and a gap on one. Two units must meet all three arcs. Two units on one term meet at most two. Two units on two spreads meet all three. A unit on a spread and one on a gap meet all three exactly when the gap lies on the arc that the spread does not. The fractions follow from $2a_y = p - r$, $2a_z = q - p$ and $2a_x = 2 + r - q$. $\square$

Lemma 14 (one sign on a line). In a frozen state with every hole oriented, on a cell of the family, $n_k(v + 8e_k) = n_k(v)$ for every vertex, and the vertices of type k on a line $\{v + 4ne_k\}$ have one sign.

Proof. By Theorem 17, n_x is a sum of weights of the chains of the axes y and z through the vertex. The translation by $8e_x$ maps each of these chains to itself. For a chain of the family $\{0, 2\}$ it is t steps of $(2, 0, 2)$ with $t \equiv 4$ modulo $2a$ and $t \equiv 0$ modulo $2c$, which has a solution because $\gcd(2a, 2c) = 4$, and the three other families are the same. So n_x has period 8 along x . A line in the direction k has fraction 0, $\frac{1}{2}$ or 1. If it is $\frac{1}{2}$ the vertices of type k are 8 apart. If it is 1, two consecutive vertices are the rods of a straight hole, which is oriented, so they have the same sign. $\square$

So $\sigma_k = \pm a_k$ on every line. With the functions of Theorem 19 put $F_{\pm} = F_{01} \pm F_{23}$, $U_{\pm} = U_{02} \pm U_{13}$ and $W_{\pm} = W_{03} \pm W_{12}$, so that on A

$$\sigma_x = U_+(y) + W_+(z), \quad \sigma_y = F_+(x) + W_-(z), \quad \sigma_z = F_-(x) + U_-(y).$$

The three pairs (U_+, W_+) , (F_+, W_-) and (F_-, U_-) are each determined by the fractions of A up to a constant added to one member and taken from the other.

Lemma 15 (the fractions of B). On B

$$\begin{aligned} 2\sigma_x &= U_+(y-1) + U_+(y+1) + U_-(y-1) - U_-(y+1) \\ &\quad + W_+(z-1) + W_+(z+1) + W_-(z-1) - W_-(z+1), \\ 2\sigma_y &= F_+(x-1) + F_+(x+1) + F_-(x-1) - F_-(x+1) \\ &\quad + W_+(z-1) - W_+(z+1) + W_-(z-1) + W_-(z+1), \\ 2\sigma_z &= F_+(x-1) - F_+(x+1) + F_-(x-1) + F_-(x+1) \\ &\quad + U_+(y-1) - U_+(y+1) + U_-(y-1) + U_-(y+1), \end{aligned}$$

and the right sides do not change when the constants are moved.

Proof. Theorem 19 gives the fractions of B as those of A with F_{01} , U_{02} and W_{03} taken one plane lower and F_{23} , U_{13} and W_{12} one plane higher. Substitute $F_{01} = \frac{1}{2}(F_+ + F_-)$, $F_{23} = \frac{1}{2}(F_+ - F_-)$ and the like. $\square$

Two remarks are used several times. If a_k on S depends on one transverse coordinate and vanishes on some plane of it, then σ_k vanishes on that plane, and the part of σ_k that depends on the other coordinate is constant. And if a function f on the planes of one sublattice has $f(t-1) - f(t+1)$ the same for every t , the sum round the circle shows that the difference is zero and f is constant.

Lemma 16 (an axis with both signs). If the axis k is two-way on S , then a_k on S depends on one transverse coordinate only.

Proof. The functions $\frac{1}{2}(a_k + \sigma_k)$ and $\frac{1}{2}(a_k - \sigma_k)$ are the fractions of the two directions on the lines of S . They are not negative, they are separable by Theorems 15 and 19, their product is zero by Lemma 14, and neither vanishes identically. By Lemma 4 both depend on the same single coordinate. $\square$

Lemma 17 (the two sublattices). The mean of σ_k over the lines of A equals its mean

over the lines of B. So an axis that is one-way on A and one-way on B, with vertices on both, is one-way.

Proof. As y runs over the odd values, $y - 1$ and $y + 1$ each run over the even values, so the mean of $U_{02}(y - 1) + U_{13}(y + 1)$ over B is the mean of U_+ over A, and the same holds for the other functions. If the axis had opposite signs on the two sublattices the two means would have opposite signs. $\square$

Lemma 18 (a case of the first kind). If S is in the case $(i)_k$, the axis k is one-way on S . If A is in the case $(i)_k$ and B carries all three types, B is in the case $(i)_k$.

Proof. Take $k = x$. The fraction a_x depends on both y and z , so by Lemma 16 the axis is not two-way on S , and it is not empty.

For the second statement let ς be the sign of the axis x on A, so that $\sigma_x = \varsigma(1 - a_y(z) - a_z(y))$ on A, and after moving a constant $U_+(y) = -\varsigma a_z(y)$ up to a constant c_1 . The fraction a_y of A depends on z and vanishes on some plane, so F_+ is constant, and a_z depends on y and vanishes on some plane, so F_- is constant. Write f_+ and f_- for the two constants. Then $\sigma_z = f_- + U_-(y)$ on A is a function $t(y)$ with $|t| = a_z$. Put $\nu_{\pm}(y) = \frac{1}{2}(a_z(y) \pm t(y))$, which take the values 0 and $\frac{1}{2}$ and have product zero.

By Lemma 15 the parts of σ_y and σ_z on B that depend on x are the constants f_+ and f_- . So on B the fraction a_y depends on z only and a_z on y only, and p is constant on B. By Lemma 13 B is in one of the cases $(i)_x$, $(ii)_y$ and $(ii)_z$.

Suppose B is in the case $(ii)_z$, so that $|\sigma_z| = \frac{1}{2}$ on every line of B. Substituting $U_+ = c_1 - \varsigma(\nu_+ + \nu_-)$ and $U_- = \nu_+ - \nu_- - f_-$ in Lemma 15 gives, on B,

$$\sigma_z = \nu_+(y + 1) - \nu_-(y - 1) \text{ if } \varsigma = 1, \quad \sigma_z = \nu_+(y - 1) - \nu_-(y + 1) \text{ if } \varsigma = -1.$$

Take $\varsigma = 1$. For every odd y exactly one of $\nu_+(y + 1)$ and $\nu_-(y - 1)$ is $\frac{1}{2}$. Let y_0 be a plane of A with $a_z(y_0) = 0$. Then $\nu_-(y_0) = 0$, so $\nu_+(y_0 + 2) = \frac{1}{2}$. If $\nu_+(y_0 + 2m) = \frac{1}{2}$ then $\nu_-(y_0 + 2m) = 0$, and so $\nu_+(y_0 + 2m + 2) = \frac{1}{2}$. Going round the circle gives $\nu_+(y_0) = \frac{1}{2}$, which contradicts $a_z(y_0) = 0$. For $\varsigma = -1$ the same argument runs in the other direction.

Suppose B is in the case $(ii)_y$, so that $|\sigma_y| = \frac{1}{2}$ on every line of B. Now $W_+(z) = -\varsigma a_y(z)$ up to a constant, and $\sigma_y = f_+ + W_-(z)$ on A is a function $t'(z)$ with $|t'| = a_y$. With $\nu'_{\pm} = \frac{1}{2}(a_y(z) \pm t'(z))$ Lemma 15 gives $\sigma_y = \nu'_+(z - 1) - \nu'_-(z + 1)$ on B if $\varsigma = 1$ and

$\nu'_+(z+1) - \nu'_-(z-1)$ if $\varsigma = -1$, and the argument round the circle of the planes z , from a plane on which a_y vanishes, gives the same contradiction. $\square$

Lemma 19 (two cases of the second kind with one axis). If A and B are both in the case (ii)_k, the axis k is one-way.

Proof. Take $k = z$. On A the fractions a_y and a_x depend on z and each vanishes on some plane, so F_+ and U_+ are constant. The same holds on B, so by Lemma 15 the functions $F_-(x-1) - F_-(x+1)$ and $U_-(y-1) - U_-(y+1)$ are constant, and then F_- and U_- are constant. So σ_z is a constant on A, of modulus $\frac{1}{2}$, and by Lemma 15 it is the same constant on B. $\square$

Lemma 20 (two cases of the second kind with different axes). On a cell of the family no solution of the type model has $a_z = \frac{1}{2}$ on every line of A in the direction z , $a_x = \frac{1}{2}$ on every line of B in the direction x , and a vertex of type y on A. So if A and B carry all three types and are both in cases of the second kind, they are in the same case.

Proof. The second statement follows from the first by a permutation of the coordinates, which maps the family to itself and carries any two different axes to z and x .

Number the classes $c_0 = (+, +, +)$, $c_1 = (+, -, -)$, $c_2 = (-, +, -)$ and $c_3 = (-, -, +)$. Each value $E_z^A(n, \sigma)$ is the value of E_z at the vertices of A of one class, the one with $\rho_y = \sigma$ and ρ_z given by n , and likewise each $E_y^A(n, \sigma)$. For a class c write $b_x^c(y, z) = E_z - E_y$ at its vertices with those coordinates. Expand b_x at a vertex (x', y', z') of B by the relation between the sublattices, and collect the eight values of A by class. At a vertex of the class $(+, +, +)$ of B this gives

$$2b_x = b_x^{c_0}(y'-1, z'-1) - b_x^{c_1}(y'+1, z'+1) - b_x^{c_2}(y'-1, z'+1) - b_x^{c_3}(y'+1, z'-1),$$

and at a vertex of the class $(-, +, -)$

$$2b_x = -b_x^{c_0}(y'-1, z'+1) - b_x^{c_1}(y'+1, z'-1) + b_x^{c_2}(y'-1, z'-1) - b_x^{c_3}(y'+1, z'+1).$$

On A every line in the direction z alternates, so $q = p + 1$ are constant, the fractions a_x and a_y depend on z , and each plane z of A has $a_x = \frac{1}{2}$ and $a_y = 0$, or $a_x = 0$ and $a_y = \frac{1}{2}$. Call the planes of the first kind Z_x and those of the second Z_y . There is a vertex of type y , so Z_y is not empty. By the rule of parity $|b_x^c(y, z)|$ is $\frac{1}{2}$ for z in Z_x and 0 for z in Z_y . On a

plane of Z_x the fraction a_y vanishes, so $E_x = E_z$ at its vertices, and in each class E_z has the same value on every plane of Z_x . So $b_x^c(y, z)$ does not depend on z in Z_x . On B every line in the direction x alternates, so $|b_x| = \frac{1}{2}$ at every vertex of B.

Two planes z_0 and $z_0 + 2$ cannot both be in Z_y , since at a vertex of B with $z' = z_0 + 1$ every term of the expansion would vanish. Take z_0 in Z_y . Then $z_0 - 2$ and $z_0 + 2$ are in Z_x . They are congruent modulo 4, and two of the four classes have vertices on them, c_0 and c_3 if they are multiples of 4 and c_1 and c_2 if not. Fix $y' \equiv 1$ modulo 4. The planes $z_0 - 1$ and $z_0 + 1$ of B have opposite ρ_z , so with this y' one has vertices of the class $(+, +, +)$ and the other of the class $(-, +, -)$. In each of the two expansions the two terms on the plane z_0 vanish, and the two others are the same two numbers u and u' in both, each $\pm \frac{1}{2}$, since b_x^c does not depend on the plane of Z_x . They enter one expansion with the same sign and the other with opposite signs. So $|u + u'| = 1$ and $|u - u'| = 1$, which is impossible. $\square$

Theorem 23. On a cell $(2a', 2b', 2c')$ with a', b' and c' pairwise coprime, every odd frozen state has an axis that is empty or one-way. If all three types occur, both sublattices are in the same one of the six cases of Lemma 13, and the axis of that case is one-way.

Proof. Suppose all three types occur. Both sublattices carry all three, and each is in one of the six cases. If A is in a case $(i)_k$, so is B, by Lemma 18, the axis k is one-way on each, and by Lemma 17 it is one-way. If B is in a case $(i)_k$ the inversion, which exchanges the sublattices and preserves the cases, gives the same. Otherwise both are in cases of the second kind, of the same axis by Lemma 20, and Lemma 19 applies. $\square$

Corollary. On these cells every frozen zero-flux state is invariant under a nonzero translation.

Proof. A state with even curl has a one-cube period by Theorem P. In a zero-flux state each axis has as many moments of one sign as of the other on each sublattice, so an axis that is one-way is empty. An odd state has therefore an empty axis, and Lemma A applies. $\square$

The proof of Theorem 23 uses Theorem 22, Theorem M' , Theorem 20, Theorems 17 and 19 and Lemma 4. It does not use K_u , Theorem 1, the walls or the diagonal law. The types enter through the six cases and through Lemma 20, and the signs through Lemmas 14 to 19. Two statements that hold in every state examined are not proved. In a case of the second kind all three axes were one-way in each of 112 sampled states, and the theorem gives one.

And the two sublattices have the same case already in the type model, by solver, while the proof above uses the signs for the cases of the first kind.

14.4 Checks

Theorems 20 to 22 were found and proved by one agent of the project, and Lemmas 13 to 20 with Theorem 23 by another. The main session recomputed both proofs from the definitions. Two separate agents then read them as checkers, each with its own programs, and found no gap. The first corrected the account of the other positions of the cage, which had named a reflection that is not a symmetry of the lattice, and the text above has the correction. The second wrote out in full the two parts of Lemmas 18 and 20 that are given here by reference to a parallel argument, and removed steps that the proofs did not need.

Theorem 20 and its consequences hold exactly on 164 sampled solutions of $(2, 2, 2)$, $(2, 4, 6)$ and $(4, 6, 10)$, and the tests fail when a sign is deliberately changed. The identity $(\star)$ holds as an identity of linear forms in the weights on five cells. The case of Theorem 21 in which no plane is integral cannot occur together with the bridgeheads, so it was tested by an enumeration of the values of the functions E that it allows, which found no exception to either conclusion. In the type model the cage with a vertex of the type of the rods on the other sublattice is unsatisfiable on four cells in all eight positions and for all three axes, with every control satisfiable.

On 784 odd states of seven cells of the family Lemmas 14, 15 and 17 hold exactly, and the second statement of Theorem 23 holds on each of the 574 of them that have all three types, 462 in a case of the first kind and 112 in one of the second. The hypotheses of Lemma 20 are unsatisfiable in the type model for all six ordered pairs of axes on five cells, with satisfiable controls. A search by solver for an odd state with three types in which the sets of constant profiles differ on the two sublattices found none on four cells.

15 A state repeated periodically, and the cubic cells of side one to three

Lemma 21. A frozen state of the cell (a, b, c) , repeated periodically, is a frozen state of the cell $(k_1 a, k_2 b, k_3 c)$ for any positive integers k_1, k_2, k_3 . It has an odd hexagon exactly when the original has, and the same set of moments. So Theorem H on the larger cell

implies Theorem H on the smaller.

Proof. The projection of the larger cell onto the smaller carries bonds to bonds of the same direction, so the repeated state obeys the ice rule. A hexagon is a closed walk of six bonds with no displacement on the lattice, so the projection carries the hexagons of the larger cell to those of the smaller, with the same directions in the same order. A hexagon of the repeated state therefore circulates exactly when its projection does, and has the curl of its projection. Every hexagon of the smaller cell is a projection. The moment at a vertex is that at its projection. An odd frozen state of the smaller cell with every axis two-way would give one of the larger. $\square$

Theorem 24. Theorem H holds on the cubic cells of side one, two and three.

Proof. By Lemma 21 and Theorem 2 at side six. $\square$

Side two is also a case of Theorem 23. The lemma gives nothing towards a proof by hand on a cubic cell. Every cell that the cubic cell of side L repeats to has extents that are multiples of L , so it has at least L chains in every pair of planes, and none has a coprime pair. It also gives nothing beyond the certificates, and the cubic cells of side seven and more need something else.

The lemma was checked exactly on saved states. The hexagons of $(6, 6, 6)$ project to hexagons of $(3, 3, 3)$ with the same axis at each vertex, and for each of 686 states of $(3, 3, 3)$ the repeated state has (T2) on every hexagon, eight times as many odd hexagons, the repeated sign field, and the same axes empty, one-way and two-way. The same holds for 54 states of $(2, 2, 2)$ repeated to $(4, 4, 4)$ and 25 repeated to $(6, 6, 6)$.

16 Three chains in every pair of planes

On the cells $(3a', 3b', 3c')$ with a', b' and c' pairwise coprime, among them $(3, 3, 3)$, $(3, 3, 6)$ and $(6, 9, 15)$, every pair of planes holds three chains of each family. The method of the last section does not prove statement (V) on them. This section records how far it goes. It reports the work of one agent of the project. The main session ran its programs again and did not recompute its proofs, and no checker has read them. The proofs are in the report named under Artefacts and are not reproduced here.

The description of a solution goes over. The three chains of a pair of planes are told apart by a label modulo 3, and the weight of a chain is the mean of its pair of planes and a

complex number multiplied by a power of $\omega = e^{2\pi i/3}$. Along a line in the direction k the indicator of the type k is

$$K_k = a_k + 2 \operatorname{Re} [\omega^s \beta_k],$$

where s is $(x + y + z)/4$ on A and advances by one at each step, and a_k and β_k are functions of the two transverse coordinates. A line has period 3 and carries one of the eight patterns of three values 0 and 1. Its fraction a_k is 0, $\frac{1}{3}$, $\frac{2}{3}$ or 1, and $3\beta_k$ is 0 or one of the six units of the ring $\mathbb{Z}[\omega]$. The rule of parity becomes a rule of colour: $3\beta_k$ is congruent to $3a_k$ modulo $1 - \omega$, where the quotient of the ring by $1 - \omega$ is the field of three elements. There are integer profiles with $3a_y = p(x) - r(z)$, $3a_z = q(y) - p(x)$ and $3a_x = 3 + r(z) - q(y)$, and the three spreads and three gaps add to 3. The data of B are those of A moved by one plane, with a factor ω on three of the six families. All of this holds exactly on 776 sampled solutions of three cells.

Of the five steps of the proof of Theorem 21, three go over in a changed form. When the line of the rods is full, the profile p of A is constant, and the lines through the rods leave one real number $\tau(x)$ free on each plane x of A. The identity of the bridgeheads becomes

$$p^B(1) - p^B(3) = 3 - 6 \Sigma(0), \quad \Sigma(0) = \tau(0) + \tau(2) + \tau(4),$$

with a sum over three consecutive planes, one of each class, in place of a difference within one class. If $\Sigma(0) = 0$ the conclusions of Theorem 21 follow. If some plane of A of a class other than that of the rods has an integral fraction, a sum over the three classes gives $\Sigma(0) = 0$. In every case $\Sigma(0)$ is 0 or $\frac{1}{3}$.

Two steps fail, and each leaves a statement that is true by solver on the six cells tried and is not proved.

Statement R. Two rods of type x four apart, a lower bridgehead of type y and the upper bridgehead on the other side not of type y make the whole line of the rods of type x .

At two chains the line has period 2 and the two rods fill it. At three it has period 3, the rods fix two positions of three, and the rods alone do not force the third.

Statement S. If the line of the rods is full, then $\Sigma(0) = 0$.

At two chains the corresponding case was closed by a count: all three profiles of B would be non-constant, and their spreads add to at most 2. At three chains the spreads add to at

most 3, and three spreads of 1 are allowed. The rule of colour gives three different residues for $p^B(1)$, $p^B(3)$ and $p^B(5)$ and no bound on the spreads of the two other profiles.

Neither statement follows from the linear programme. Each follows from it when a few lines are required to be integral, one line for R and four for S, and the linear parts are then certificates for each cell, of 20 to 47 rows, in which no structure was found. The one for S needs fifteen layers along the rod axis and the whole transverse torus. With both statements, (V) holds on these cells in every position of the cage. Theorem H would need K_u in addition, or a counterpart of Theorem 23, and neither was attempted.

The method used that a function on three points with values 0 and 1 has one complex coefficient apart from its mean. With five chains or more a line has several, and the condition that its values are 0 and 1 couples them, so the rule of colour has no counterpart of the same kind. The work gives no sign that the method of the last section reaches the cubic cells of general side.

17 Theorem U

Theorem U is proved on the cells with a coprime pair of extents and on the cells with two chains in every pair of planes, and it is certified on the cubic cells of side four to eight and on the cells $(3, 3, 9)$, $(4, 4, 8)$, $(3, 3, 12)$ and $(6, 6, 9)$. On every other cell it is open, and since 30 September 2026 it is the one open statement between the record and Theorem H, and the periodicity of frozen zero-flux states, on every cell (Theorem 1). This section and the next are about those cells. They were written before the sections on the two families, and they record what computation has found about the statement and which routes it has closed.

The statement has several forms, equivalent in a frozen state by the results above.

1. An odd state has no charged hole.
2. In an odd state two consecutive vertices of type k on a k -line have the same sign.
3. A state with an unoriented cage has even curl.
4. (V) In a state with the four vertices of Theorem U' , the sublattice of the bridgeheads has no vertex of type k . Even curl then follows from Theorem M' .

The solver decides it in seconds, 0.2 s at side three and 22 s at side six, and the pattern

forces a great deal. Every vertex of the other sublattice is forced not to be of type k , the line of the two rods is forced to be entirely of type k , and the two planes of bridgeheads normal to k on either side of the hole are forced to be entirely of type a and entirely of type b . States with even curl and unoriented holes exist, among them states with all six directions. In them the unoriented holes sit in pairs of opposite charge on complete lines of rods.

The following facts were established by computation on the cells named in the reports, and each closes a route to a proof.

1. On the cubic cell, with (T₂) imposed only on the hexagons within R layers of the hole along the rod axis, the pattern forces nothing until $R = 2L - 2$ of the $2L$ available, and the transverse torus is needed in full. No ball, tube, slab or box short of these suffices. The deletion-minimal sets of hexagons contain only a fifth to a third of the lattice, but they meet every plane in every direction. This does not mean that the statement needs the circle of the rod axis, and the symmetric window is not the smallest. On a cell with extents (a, b, c) and the rod axis of length a , put $g = \gcd(b, c)$. For $g \geq 4$ the smallest windows of layers $[-p, q]$ about the hole that force the conclusion are $(p, q) = (3, 2g + 1)$, $(2g - 2, 2g - 2)$ and $(2g + 1, 3)$, whatever the value of a . For $g \leq 2$ the window $(3, 3)$ suffices, and at $g = 3$ the smallest are $(3, 7)$, $(4, 5)$ and their reflections. The cubic cell has $g = L$, so there the symmetric threshold $2L - 2$ looks like most of the circle, but the window $(3, 2L + 1)$, which is $2L + 5$ of the $4L$ grids, suffices at side five, six and seven. So (V) is a statement about a slab of $2g + 5$ grids over the closed transverse torus, and no window in the transverse directions suffices on any cell tried: removing the hexagons through a single transverse grid leaves the negation satisfiable. (Added on 27 September 2026 after the note was first issued. Solver, on twenty transverse tori with g from 1 to 8, confirmed by an independent checker with its own implementation, with and without the sign field. The corners at side four and five are in the report of 22 September on the core of (V), and the dependence on g was found on 16 and 17 September for an earlier form of the statement.)
2. The next cage along the line of the rods is forced to be straight and its bridgeheads are free. No other hole of either sublattice is forced to be oriented or to be unoriented. Charges are neutral on every plane of holes, which is the flux identity, so an unoriented hole has

partners, and none of them is determined.

3. When the types are relaxed to non-negative reals subject to the linear equations, the largest possible amount of type k on the plane of bridgeheads is $26/3$, $178/9$ and $299/9$ at side three, four and five. The cage laws span the linear consequences of (T₂), so no weighted sum of laws proves the theorem on these cells. Adding the signs, the bond rule and Lemma 3 lowers these numbers by less than one. On cells whose two transverse extents are coprime the largest amount is zero, on every such cell tried, so there (V) does follow from a weighted sum of laws, and the first case that needs more is $g = 2$. The weighted sum needs more room than the solver does. On a long rod axis it needs a slab that grows with the torus, of 3, 7, 9 and 13 layers either side for the tori (3, 4), (4, 5), (5, 6) and (7, 8), and on a short one it uses the circle of the rod axis. (Added on 27 September 2026. Computed in floating point by two implementations, and verified in exact arithmetic by the independent checker on the cells (6, 3, 4), (6, 4, 5) and (2, 4, 5).)
4. Every window that fails without the sign field fails with it.
5. With the indicators of one type on one sublattice required to be 0 or 1 and everything else relaxed, the optimum is zero. At side three the charges of the holes of one sublattice suffice. A case split on six vertices at side three, and on nine at side four, leaves linear programmes that are all infeasible.
6. If the line law at gap one is assumed as an axiom, the reach along the rod axis falls from $2L - 2$ to $2\lceil L/3 \rceil + 1$. The proofs of the line laws given here assume orientation, so this is circular.
7. The obstruction is the closure of a sheet, and no quantity that has been tried carries it. One grid short of the window $(3, 2g + 1)$ the negation of (V), in its four-literal form, has exactly two solutions for the set of vertices of type x , at every g from 3 to 7, given by one formula in the labels of the chains of axis x modulo $4g$: a sheet on a plane normal to a bond direction through the rod, advancing one chain label per cube, with two defects that ride on a chain of axis y and a chain of axis z . The two solutions are the two senses of the advance. On every shorter window the solutions are many and dense, and no Fourier coefficient of the types on the chain labels, no count on the chains or on their cells and no winding of the set of type x is defined on all of them, or constrained on those on which it is defined; the only exact relations between grids are linear laws. Opening the transverse

torus does not remove g : on a cylinder the reach is set by the number of chain labels of the circle that is left closed, and on the plane the statement is false. On a window the integer solutions of the negation span an affine space much smaller than the real solutions, of dimension 90 against 210, 152 against 333 and 234 against 484 on the cells $(12, 4, 4)$, $(12, 5, 5)$ and $(12, 6, 6)$ with the windows $[-3, 2g - 2]$, cut out by variables that are fixed in every solution and by equations between two variables, which depend on the window. (Added on 30 September 2026. Solver and exact computation by two agents on 28 and 29 September, a sample run again by the main session; no independent checker.)

8. Three facts of 30 September 2026, each by solver with controls, a sample run again by the main session. First, the zero-flux sector does not shorten the statement: with $W = 0$ imposed on the torus the least one-sided window that forces (V) is $[-3, 2g]$ in place of $[-3, 2g + 1]$ at $g = 4, 5$ and 6 , so the sector removes exactly the grid of the two rigid solutions of item 7, which carry flux, and nothing more. Second, the statement that the two rods fill their whole line (Statement R of the section on three chains) has on the torus exactly the reach of (V), one-sided $\max(6, 2g + 1)$ and symmetric $\max(6, 2g - 2)$, and follows from (V) in two lines (Lemma 6 and law P^x); but it is strictly weaker, since it survives the removal of a transverse grid along one axis, where (V) does not, and its first step, that the third rod position is of the type of the rods, is forced inside a box over one closed transverse circle, $x \in [-3, 2b + 1]$ and $|z| \leq 2b - 1$ with the y -circle of the cell (a, b, c) whole, whatever a, c and $\gcd(b, c)$; DRAT certificates of this box lemma exist for $b = 3$ to 6 . Third, in the heights of (C) the statement R has no linear content: the difference $h_y - h_z$ at the j th position of the line depends on j only through j modulo $\gcd(a, c)$ and modulo $\gcd(a, b)$, so on a cubic cell the parametrisation relates no two positions, which is the feasible linear programme of the section on three chains said in another language; R, like (V), is an integrality statement. A correction to the record made the same day: the weights of (C) can be taken integral on 19 of the 637 odd states of the cubic cell of side three and 20 of the 720 of side four, and on none of 60 sampled solutions of the type model, with least denominators from 27 to 96 otherwise; the report of 10 September that says they can be taken integral “on every state examined” had examined those 19.
9. Facts of 1 October 2026, each by solver with controls or by exact computation, a sample run again by the main session, which close the plan that item 8 suggested (Statement R

first, then a monodromy in the weights of (C) along the rod axis, then an integrality step). First, Statement R assumed on the whole rod line shortens the one-sided reach of (V) from $2g + 1$ to $2g - 1$ on five cells with g from 4 to 8, and the single literal that the third rod position is of type k gives the same $2g - 1$ on every cell: R contributes nothing else inside a window, and (V) given R is false on the cylinder with a transverse grid removed along either axis, so R supplies neither transverse circle. One grid short of the threshold the solutions of the negation given R are many and are not label classes of either family of chains of the rod axis. Second, on the grid of the bridgeheads of every cell whose two transverse extents divide the extent along the rod axis, every cubic cell among them, each chain of the two transverse axes meets the grid once, and the parametrisation (C) restricted to the grid is onto: it imposes no linear condition on the type of the rods' axis on a grid, the pair of labels of the two transverse chains through a vertex of the grid is never a product on any cell, and Lemma 8 has nothing to act on. Third, the weights of (C) seen from successive grids along the rod axis are sheared by one label per cube and return after the least common multiple of $\gcd(a, c)$ and $\gcd(a, b)$ cubes, which is 4 on $(12, 8, 8)$, where (V) given R needs 15 grids; so the g of the reach law is the modulus of the labels $y - z$ and $y + z$ of the chains of the rod axis on the grids normal to it, and no monodromy in the weights of (C) along the rod axis is the mechanism on every cell. Any proof of U must be integral from the first line, and the integral object is the set of those labels, of both families, on each pair of grids. Fourth, Theorem 9's method applied to the box lemma of Statement R: the box lemma has no linear content, every integral family of vertices that closes its linear programme at one b fails at the next for $b = 3$ to 5, and the exact certificates of its branches are of two kinds, local ones that are identical for every b and are finite lemmas on two cages, and global ones that use the whole closed circle and grow with b , in which no rule was found. Fifth, the search of this date for a counterexample to U on the cheapest open cells, recorded in the section on the cubic cells, found none, with no near miss.

Theorem U belongs to a family of statements of the project in which a pattern on a few vertices forces a complete plane. One member of the family has a proof, Theorem M' with the sheet theorem behind it, and that proof is two integrations round the transverse torus followed by the integrality step of Lemma 4.

18 Ideas for Theorem U

This section is of 27 September 2026 and is kept as it was written, apart from the notes of later dates that it carries. None of the ideas after the first subsection has been tried unless it says so. The first subsection led to the three sections on the cells with a coprime pair, on the signed moments and on the cells with two chains. Those sections did not come from the ideas that follow it. They came from looking for quantities that the geometry makes separable, and not for certificates or case splits. The second idea, an induction in the number of chains, is what the section on three chains tried, and it stops there.

18.1 Cells that are not cubic

This subsection was the first to separate the length of the rod axis from the size of the transverse torus. Theorem 14 has since settled Theorem H on every cell with a coprime pair of extents, and Theorem 23 on the cells with two chains in every pair of planes, so what this subsection says about Theorem H on the easier members is superseded. Theorems 10 and 11 and their proofs stand. Theorem 11 reaches holes of cells that neither of the two later theorems covers, such as those of $(15, 6, 10)$ with rods along the first axis. The cubic cell of side L is the member $g = L$ of a family of cells (a, b, c) indexed by $g = \gcd(b, c)$, and in that member the height of the slab, the number of parallel chains in a layer and the depth of the case split all grow together. The family has easier members. At $g = 1$ the statement is linear and is proved below. At $g = 2$ the statement needs integrality and lives on seven layers, on tori of any size. The relative W_r of the fifth idea below is not of this kind: its reach along the normal of its plane is twice the extent of the cell along the axis of the type of its bridgeheads, with no dependence on a greatest common divisor, so it has no easier members. The induction of the second idea is then an induction on g and not along a line of holes. A state on a cubic cell lifts only to cells whose extents are multiples of L , so a proof on the easier members does not give the cubic cell directly. The measurements and an assessment of each idea below are in the report named under Artefacts.

The reduction itself does not need the cell to be cubic. An audit of every step on the cell (a, b, c) found that Theorem 1 holds whenever K_u holds for uniform chains of all three axes. A second reader confirmed this, and corrected the audit's conclusion that the

recorded scope of K_u gives it when each extent is at least four: as the first section says, the certificates give K_u for chains of an axis only when the two transverse extents have greatest common divisor at most two. The proofs of this note go through with two changes of wording: in Theorem 5 the sum is over the a_k holes of a line, and a chain of axis k has four times the least common multiple of the two other extents as its number of vertices. In the proof of Theorem M' the step that says a band of edges meets a plane of kinks exactly once is true only when the two transverse extents are equal, and what the proof uses is that it meets the plane at least once, which holds on every cell. The scope of K_u is not symmetric: its three is the extent along the axis of the chain, and the wall of Theorem 7 fixes that axis, so cells with an extent of three are not covered. On a cell whose three extents are pairwise coprime no frozen state is odd, by Theorem R1 of the project's reports, and Theorem H holds there with no conjecture. (Audit by an agent, with solver checks on nine cells, a sample of them run again by the main loop, and read by a second agent, which confirmed Theorem M' on every cell by both of its proofs and prefers the one that does not count crossings.)

Theorem 10 (the coprime case; added on 27 September 2026). On a cell with extents (a, b, c) and $\gcd(b, c) = 1$, a frozen state with an unoriented hole whose rods lie along x has even curl. So in an odd state of such a cell every hole with rods along x is oriented.

The proof uses the parametrisation of the project's working note, proved there for every cell. Attach a real weight $f(C)$ to every chain C , and for a vertex v and an axis k let $F_k(v)$ be the sum of the weights of the two chains of axis k through v . Then every real solution of the one-hot equations and of (T2), and in particular the indicators $Y = [\tau = y]$ and $Z = [\tau = z]$ of a frozen state, has the form

$$Y(v) = F_x(v) - F_z(v) + \sigma_v \alpha, \quad Z(v) = F_y(v) - F_x(v) + \sigma_v \beta,$$

with $\sigma_v = \pm 1$ on A and B and constants α, β . A chain of axis x lies in two neighbouring planes $x = \text{const}$, its vertices of one sublattice are $p + n(0, 2, \pm 2)$, and it closes after $2bc/g$ steps, so each such pair of planes holds g chains of its family. When $g = 1$ it holds one, and $F_x(v) = \Phi(x_v)$ depends only on the plane of v .

Proof. Let the hole be O , with rods $v = O - 2e_x$ and $v' = O + 2e_x$ of type x , lower bridgeheads $\beta = v + d_0$ and $\beta' = v + d_1$ of type y , and upper bridgeheads $\gamma = v' + d_2$

and $\gamma' = v' + d_3$ of type z , which is the unoriented cage up to the symmetries that fix the axis x . Write $\Phi(n)$ for Φ on the plane $x = x_O + n$. The chains of axis z are those of the pairs of directions $\{d_0, d_3\}$ and $\{d_1, d_2\}$. The vertex β lies on the first through v and γ' on the first through v' , and β and γ' lie on one chain of the second kind, through $O + 2e_z$. Likewise β' and γ lie on the chains of the second kind through v and v' and share one of the first kind, through $O - 2e_z$. Subtracting the formula for Y at γ' from that at β , and at γ from that at β' , and adding,

$$2 = 2(\Phi(-1) - \Phi(1)) - (F_z(v) - F_z(v')).$$

The rods have $Y = 0$, so $F_z(v) - F_z(v') = \Phi(-2) - \Phi(2)$, and

$$\Phi(-1) - \Phi(1) = 1 + \frac{1}{2}(\Phi(-2) - \Phi(2)).$$

Now choose an integer m with $\gcd(a, b) \mid m$ and $\gcd(a, c) \mid m + 1$, which exists because the two divisors are coprime, and put $P = v' + 4me_x$. The vertices of A on the chains of axis z through v' are $v' + n(2, \pm 2, 0)$, and P is among them, for both signs, exactly when $\gcd(a, b)$ divides m . In the same way $P = v + 4(m + 1)e_x$ lies on both chains of axis y through v . So $F_z(P) = F_z(v')$ and $F_y(P) = F_y(v)$, and since $Y(v') = Z(v) = 0$,

$$Y(P) + Z(P) = (\Phi(x_P) - \Phi(2)) + (\Phi(-2) - \Phi(x_P)) = \Phi(-2) - \Phi(2).$$

Finally add the formula for Y over a plane of B. A chain of axis z advances by 2 in x at every step and closes after a multiple of $2a$ steps, so it meets every plane of B equally often, and the sum of F_z over a plane of B is the same for every such plane. With $N = 2bc$ the number of vertices of a plane and $n_y(n)$ the number of type y on the plane $x_O + n$,

$$n_y(-1) - n_y(1) = N(\Phi(-1) - \Phi(1)) = N + \frac{N}{2}(Y(P) + Z(P)) \geq N.$$

Since $n_y(-1) \leq N$ and $n_y(1) \geq 0$, the plane of the lower bridgeheads is entirely of type y . This is (V). By Lemma 6 sublattice B has no vertex of type x , A has one, and Theorem M' gives even curl. $\square$

The argument is linear. It uses that the types are non-negative and sum to one, and not

that they are integers, which is the content of the third item of the list above for these cells. It goes once round the circle of the rod axis, through the vertex P and through the sums over planes, so it is a proof on the torus and says nothing about a slab. It proves Theorem U for the holes of one axis at a time. All the holes of a cell are covered only when the three extents are pairwise coprime, where there are no odd states at all, so the theorem gives no new case of Theorem H. It is the first statement of this class with a proof. The identity behind it, that the number of vertices not of type y on the lower plane, plus the number of type y on the upper plane, plus $\frac{N}{2}(Y(P) + Z(P))$, is zero, was verified in exact arithmetic on thirty-five cells by the agent that found it and by an independent checker with its own implementation, which also read the proof and confirmed it. At $g = 2$ the same identity fails for every choice of P .

Theorem 11 (greatest common divisor two, odd length; added on 27 September 2026). On a cell with extents (a, b, c) , $\gcd(b, c) = 2$ and a odd, a frozen state with an unoriented hole whose rods lie along x has even curl.

At $g = 2$ a pair of neighbouring planes holds two chains of axis x of its family, and F_x is no longer a function of the plane. Since b and c are even, the periods $4b$ and $4c$ are multiples of 8, so $y - z$ and $y + z$ are defined modulo 8 on the cell. A chain of the family $\{d_0, d_1\}$ has steps $(0, 2, 2)$ and constant $y - z$, and one of the family $\{d_2, d_3\}$ has steps $(0, 2, -2)$ and constant $y + z$. On a plane the residue takes two values, exchanged by the translation $(0, 4, 0)$, so each residue class has bc vertices, which is the length of a chain. The two chains of a pair of planes are therefore the two residue classes.

Proof. Notation as in Theorem 10, with $O = (2, 0, 0)$. Let A_0 be the chain of $\{d_0, d_1\}$ through v . It contains β and β' , and its vertices on the plane $x = 1$ are the half H_1 of that plane with $y - z \equiv 0$. Let B_3 be the chain of $\{d_2, d_3\}$ through v' . It contains γ and γ' , and meets the plane $x = 3$ in the half H_3 with $y + z \equiv 0$. The other chains of axis x through β and β' are the two chains of the planes 1 and 2, with $y + z \equiv 2$ and 6, and those through γ and γ' are the two chains of the planes 2 and 3. Write S_1 and S_2 for the sums of the weights of these two pairs. The first step of the proof of Theorem 10 now reads

$$2(f(A_0) - f(B_3)) + S_1 - S_2 = 2 + D, \quad D = F_x(v) - F_x(v'),$$

and the second, in which the weights of the chains of axis x through P cancel, gives

$D = Y(P) + Z(P)$ unchanged. Here $\gcd(a, b)$ and $\gcd(a, c)$ are odd and coprime, because a is odd and $\gcd(b, c) = 2$, so P exists.

Two counts are needed. Each of the two chains of the planes 1 and 2 has its vertices of the plane 1 alternately in H_1 and outside it, because a step changes $y - z$ by 4, and it has an even number bc of them, so half lie in H_1 . And a chain of axis y or z has as many vertices in H_1 as in the other half of the plane $x = 1$. For a chain of axis z the vertices of B are $u + n(2, \pm 2, 0)$, those on one plane are $2a$ steps apart, and there are $b/\gcd(a, b)$ of them, an even number. Between consecutive ones y changes by $4a$, which is 4 modulo 8 because a is odd, so they lie alternately in the two halves. The same holds for a chain of axis y , with c and z , and on the plane $x = 3$ with H_3 .

Adding the formula for Y over H_1 and over H_3 , the sums of F_z are half of the sums over the planes, which are equal, and

$$\sum_{H_1} Y - \sum_{H_3} Y = \frac{N}{4} \left[2(f(A_0) - f(B_3)) + S_1 - S_2 \right] = \frac{N}{2} + \frac{N}{4} (Y(P) + Z(P)).$$

Since H_1 has $N/2$ vertices, every vertex of H_1 is of type y . On B the indicator of type x is $1 - Y - Z = 1 - F_y + F_z + \alpha + \beta$, in which F_x does not occur, so by the second count the number of vertices of type x on the plane $x = 1$ is twice the number in H_1 , which is zero. By Lemma 6 sublattice B has no vertex of type x , and Theorem M' gives even curl. $\square$

The parity of a enters through the second count. In general the difference between the two halves of a plane, for the chains of axis y and z , changes sign from a plane to the next plane of the same kind, four layers on, so round the circle it is multiplied by $(-1)^a$ and vanishes when a is odd. When a is even it survives, the vertex P does not exist, and the relaxation to real types has solutions with type x on the lower plane, on the whole cell and on every slab. The proof gives less than (V): the other half of the lower plane is not forced to be of type y by any weighted sum of laws. The theorem was found by an agent, read by the main loop, and confirmed by an independent checker, and its identity was verified in exact arithmetic on thirty-three cells by two implementations. At $g = 3$ and 4 with a odd, and at $g = 2$ with a even, the relaxation has solutions and integrality is needed. No integral statement that would replace the second count has been found. The obvious one, that the vertices of type x of a plane are divided equally between its halves, is false in

frozen states with a even.

Theorems 10 and 11 together give Theorem U for every hole of a cell in which one extent is odd and coprime to the other two, and those two are even with greatest common divisor two. On the cell $(5, 4, 6)$ of this kind, however, no odd state has a vertex of type x (solver, by a split into five cases), so every odd state lacks a type and Theorem H holds for it without any argument. The two theorems are proofs of statements of the class of (V), and they give no case of Theorem H that needed them. Theorem H on these cells has a shorter proof. When two of the three pairs of extents are coprime, Theorem R1 puts the two rods of every corner on a chain of the third axis, so every corner has that axis as the one it does not use. The project's Theorem C of 24 September, which holds on every cell with the length of a chain in place of $4L$, then makes the rods of the trail of corners two complete uniform chains, and K_u for chains of that one axis gives a missing direction. This needs K_u where its certificates reach, that is, when the third pair of extents has greatest common divisor at most two.

18.2 A closed form for the branch certificates

The diagonal law stood as a family of machine-found certificates, one for each separation, until the dual solutions were made canonical, by minimising the L^1 norm and imposing the symmetries of the pair, and the multipliers tabulated against the local position of each equation relative to the diamond. The closed form of Theorem 9 then had four rules. Reading the raw certificates had found nothing. The corresponding objects exist for Theorem U, where after a case split on a few vertices every branch is an infeasible linear programme. The differences are that the support is the whole torus, so the closed form would be written in terms of L and of position relative to the planes through the hole, and that the number of branches grows with L . The first computation is to find the thinnest family of indicators whose integrality closes the programme at sides four, five and six, as was done for a neighbouring statement where one line of $2L$ indicators suffices and the split at side four is on a single vertex. The second is to make the branch certificates canonical and tabulate them. A family of certificates whose multipliers depend only on a bounded signature would be a proof.

18.3 An induction along one line

For a k -free plane with a hole whose bridgeheads disagree, Theorem 5 turns repetition into the conclusion. If the disagreement recurs at every hole of one line of holes, two neighbouring lines of the other sublattice have counts L and o , separability forces two complete planes, and Lemma 6 and Theorem M' finish. The step from one hole of the line to the next is an identity from P^k , and it fails only if one named vertex has type k . That vertex is not locally controlled, so the induction is not local. It does give the case split its shape, one vertex for each step along the line, and suggests that the branches of the first idea should be ordered along that line.

18.4 Corners and charges as two kinds of defect

A corner is a hole where a loop of rods turns, and a charged hole is one where its direction reverses. Theorem U says that no frozen state has both. The two live on the same lattice of holes, the charges on its vertices and the odd hexagons on its edges, and every linear relation between them is excluded by the computations above. An identity that is quadratic in the indicators, a product of a count of corners and a count of charges, is not excluded, and the fact that integrality of one family suffices is what such an identity would look like to a linear programme. One experiment is to relax the two indicators separately and find the largest product of the two counts that the relaxation allows on small cells.

18.5 The weaker statement that the reduction needs

Theorem 1 uses orientation at every hole, but it is applied only to states in which all three axes are two-way. It would be enough to prove Theorem U for those. The extra hypothesis is a disjunction over the torus and is unlikely to shorten any window, but it changes the target. In the structure that the unoriented cage forces, one sublattice has no vertex of type k , so both signs of k must occur on the other. Whether the forced structure is compatible with both signs of all three axes is a finite question on each cell and has not been asked.

This idea is withdrawn (27 September 2026). Given Theorem 1 and K_u , the statement restricted to states with three two-way axes is equivalent to Theorem H, because H says that no odd state satisfies its hypothesis, so it is no easier than H. The question in the last

sentence is also answered by the section on Theorem U, which records that states of even curl with unoriented holes and all six directions exist.

18.6 The relatives without oddness

Two statements close to Theorem U hold in every frozen state, with no hypothesis of oddness, by solver at sides three, four and five. A rod with its two bridgeheads of a second type on one side forces a free plane through the hole, and a pair of vertices of type k with opposite signs and a common neighbour forces a free plane through the line of that neighbour. They have the same reach as (V) and the same resistance to counting. They are simpler in one respect, since a frozen state is either odd or a sum of three one-dimensional profiles, and in the second case the types are explicit functions of the profiles. A proof of one of them in the odd case, without orientation, would be the first proof of a statement of this class.

18.7 Further cells

Theorem 9 removes two of the three certificates from the proof of H at a given side, leaving Theorem U' . Its proofs grow from 40 MB at side four to 350 MB at side six and its solving time from 2 s to 22 s, so sides seven to ten are within reach of a workstation. This extends Theorem 2 and proves nothing uniform. By Lemma 21 a certificate at one side gives Theorem H at every side that divides it, and at no other, so each further side needs a certificate at that side or at a multiple of it. The first step of K_u is already certified at sides seven and eight, so Theorem U' at those sides would give Theorem H there. (Done on 1 October 2026, as a search for a counterexample that found none: Theorem U' is certified at sides seven and eight and on four cells that are not cubic, with proofs of 0.9 to 2.9 GB; see the section on the cubic cells.)

19 Relation to the earlier reductions

The earlier note wrote H as two statements, that an odd state has a chain without kinks and that such a chain forces a direction to be absent. The second was certified for uniform chains as K_u , and the first was proved by hand on 24 and 25 September 2026 for most

of a corpus of odd states, through the trails of odd hexagons and their runs. The chains produced there are uniform, so those states were already cases of H. The statements that remained from that work, on the runs of a trail, on three free planes, on the bridgeheads at a free plane and on the existence of a free plane on a two-way axis, are no longer needed. The last is a consequence of Theorem 7 and the third of Lemma 5, in odd states and given Theorem U.

Theorems 14 and 23 are a third route, which does not pass through a chain without kinks. They use the parametrisation of the types by weights on the chains, which the working note proved on 17 September 2026, and its counterpart for the signs. The proof of Theorem 1 given on 30 September belongs to the same route: it takes from the parametrisation only Theorem 15, and it makes the chain without kinks unnecessary altogether, since the axis it produces is one-way by the law for neighbouring lines and not by K_u . One lemma of that date, on a function of two coordinates that is a sum of functions of three linear forms, was stated without a hypothesis that it needs, and Theorem R1 was proved from it. Lemma 9 states it with the hypothesis and Lemma 10 supplies the hypothesis, so Theorem R1 stands.

20 Related work

The Coulomb phase of spin ice and its emergent gauge field are reviewed by Henley (2010) and Castelnovo, Moessner, and Sondhi (2012). The flux identity used here, that the net moment through a plane is the same for every parallel plane, is the conservation law of that field. The $\langle 110 \rangle$ chains are the strings of the Kasteleyn transition in a $[100]$ field (Jaubert et al. 2008). Separable height functions and the six-vertex rule that appears at a free plane belong to the theory of the six-vertex model (Duminil-Copin et al. 2024). The literature on fully packed loops has not been searched for the structure of this note, in which two interpenetrating systems of loops are coupled through the hexagons, and a search should precede any claim of novelty.

21 Certification

The results of this note have four kinds of standing.

Proved by an argument written out here: law C, law P^k , Lemmas 1, 2, 4, 5 and 6, Theorems

4 to 8 and Theorem 1 in both its forms; Lemmas 8 to 25 and Theorems 12 to 23. Each was read by a separate agent as checker, with its own programs, and no gap was found, except Lemma 21, which was written by the main session, tested exactly on repeated states and read by no checker. Lemmas 22 to 25 and the proof of Theorem 1 of 30 September were written by one agent, re-derived by the main session from the theorems they cite before their scripts were read, and then read by an independent checker, which confirmed them with corrections of wording that are in the text; their one prediction about odd states was tested on the corpus with no exception. The table gives what each group rests on from outside this note.

results	rest on
Lemmas 22 to 25, Theorem 1	Theorem M' ; Lemma 3 through Theorems 3 to 6; the parametrisation (C) through Theorem 15
Lemmas 8 to 10, Theorems 12, 13 and 15 Theorem 14	the parametrisation (C) Theorem 12 and Theorem M' ; on cells with two coprime pairs that share an extent, Theorem 13 alone
Lemmas 11 and 12, Theorems 16 to 19 Theorems 20 and 21 Theorem 22	Lemma 3 the parametrisation (C) Theorem 21 and Theorem M'
Lemmas 13 to 20, Theorem 23	Theorems 17, 19, 20 and 22, Lemma 4 and Theorem M'
Lemma 21	nothing
Theorem 24	Lemma 21 and Theorem 2

Theorem M' and the parametrisation (C) are from the project's earlier record. Theorem M' has two proofs, with lemmas on four hexagons that carry DRAT certificates, and one second reader. The parametrisation was proved by a symbolic computation and confirmed by exact ranks on cells; on 1 October 2026 an independent checker, with its own model of the hexagons and its own stratification of the unit torus, reproduced the proof for every cell (the same 68 components) and the exact ranks on twelve cells, and closed one

expository gap in the companion kernel result. The section on three chains in every pair of planes reports work that has had no checker, and states no theorem.

Proved by an argument that rests on a finite enumeration of the size of one or two cages: Lemma 3 (1,200 patterns) and Lemma 7 (6,192 patterns), each run by at least two independent programs, and Theorem 9, whose finite parts are the fifty intersections of the tetrahedron and whose identity was also verified exactly for separations up to 101.

Certified by DRAT proofs checked by drat-trim: Theorem U' and hence Theorems U and 2 at sides four to eight and on the cells $(3, 3, 9)$, $(4, 4, 8)$, $(3, 3, 12)$ and $(6, 6, 9)$, and with Lemma 21 Theorem 24. The 114 formulas of 27 September and the 155 of 1 October, with the proofs under 50 MB, are regenerated by the scripts named below. Their SHA-256 sums and verdicts are in the repository, and the files themselves, 738 MB and 2.9 GB, are not; the proofs of 1 October over 50 MB were deleted after checking and are recorded by hash and size. Also certified, and since 30 September no longer steps of any proof here: the first step of K_u on the whole slab at sides four to eight, and the box lemma of Statement R for $b = 3$ to 6.

Taken from the project's earlier record and, on 1 October 2026, read again by an independent checker with its own programs: Lemma L and Lemma A are confirmed on every cell (Lemma A needs one line the earlier note leaves out, that $(0, 2g, 2g)$ with $g = \gcd(b, c)$ is a nonzero element of its lattice of translations); Lemma 21 is confirmed by an exact re-test on eight pairs of cells; Theorem P is confirmed with the hypothesis that every extent is at least two, which the earlier note states and the first draft of this note dropped, and the chain from Theorem H to periodicity is corrected accordingly, with the thin cells as witnesses that the hypothesis is needed. What remains unread in Theorem P is the proof of its first step (every even-curl state is a slab sum): its two linear-algebra steps, the completeness of the chain parametrisation and the kernel of the match operator, were written by agents on 17 September 2026, re-run by the main session, and have had no checker; the checker of 1 October read the rank lemma of the second and found no gap, and did not read the first. In the earlier form of Theorem 1 only, K_u was also taken from the record; it was found to have a gap on cubic cells when a checker did read it, as the first section says, and one lemma of the record was found to be false as it was stated.

The proofs were written by language-model agents working from the project's reports,

each with a separate agent as checker that was asked to find a gap, wrote its own programs from the definitions and did not use the author's. The checkers confirmed the reduction and the certificate of Theorem 9 with corrections to a citation, to the orientation of one equation and to one bound, and found no gap. Every statement was also tested on the corpus of 1,357 odd frozen states and on 186 further states at sides five and six. Two statements that the corpus supported without exception were refuted during the same work by solver witnesses, because the corpus contains odd states only, and the note relies on the corpus for no statement that could fail in a state of even curl. No human has yet read the proofs line by line.

In the work of 27 and 28 September 2026 the checkers found a step missing from K_u , a lemma stated without a hypothesis it needs, a reflection named as a symmetry that is not one, and a claim that a proof used no certificate when a theorem it cited does. Each is corrected in the text. They are recorded here because they show both that the checks find errors and that errors of this kind are common in the work.

22 History of this note

The note was first issued on 27 September 2026 with the reduction of Theorem H to Theorem U and K_u , Theorem 2, and the sections on Theorem U and on ideas for it. It was amended the same day: the scope of the first step of K_u was corrected, the first and third closed routes for Theorem U were revised, the subsection on cells that are not cubic was added with Theorems 10 and 11, and the fourth idea was withdrawn. On 28 September 2026 four sections were added, on cells with a coprime pair of extents, on the signed moments, on cells with two chains in every pair of planes, and on a state repeated periodically, with a section on three chains in every pair of planes. The title, the abstract and the first section were rewritten to state what is now proved, and the note was given this history in place of dated remarks in the abstract. On 30 September 2026 the seventh item of the section on Theorem U was added, recording the two rounds of 28 and 29 September on the closure of the sheet, which found no proof. Later the same day Theorem 1 was given its present form, with Lemmas 22 to 25, and K_u ceased to be a step of the reduction; the earlier proof is kept under its own heading, the eighth item of the section on Theorem U records the day's three facts about (V) and Statement R and a correction on the integrality of the weights,

the title, abstract, first section and certification were revised accordingly, and the PDF was rebuilt. On 1 October 2026 six agents ran. The chain from Theorem H to periodicity was read by two independent checkers: Lemma L, Lemma A and Lemma 21 confirmed on every cell, Theorem P corrected to require every extent at least two (the thin cells are counterexamples to the conclusion without it), and the two linear-algebra steps of its first step, the parametrisation (C) among them, confirmed as proofs for every cell by an independent stratification. The plan of 30 September for Theorem U was closed by the facts of the ninth item of its section, and the closed-form certificate of the first step of K_u was confirmed. A search for a counterexample to Theorem U on the cheapest open cells found none and left Theorem U' certified at sides seven and eight and on four cells that are not cubic, so that Theorem 2 now reaches side eight. The abstract, the first section, the section on the cubic cells, the section on Theorem U and the certification were revised, and the PDF rebuilt.

23 Artefacts

- docs/reports/2026-09-27-synthesis.md: the map of the argument and the record of the checks.
- docs/reports/2026-09-27-opus-rphand.md, -lines.md, -cone.md: the reduction, the line law and the certificate of Theorem 9, with the checkers' reports -check.md and -check2.md.
- docs/reports/2026-09-27-opus-cert.md: the certificates. Hashes and verdicts in experiments/08_frozen_structure/scratch_h/certs/h28x_*; regenerate with h282_certify.py.
- docs/reports/2026-09-22-orient.md, -ucore.md, -uhand.md, 2026-09-27-opus-uwall.md and -vsign.md: the record on Theorem U.
- docs/reports/2026-09-27-aniso.md: (V) on cells that are not cubic, and the assessment of the ideas. Scripts h330_aniso.py and h331_aniso_lp.py.
- docs/reports/2026-09-28-attack-proposal.md and 2026-09-28-opus-check7.md: the section on cells with a coprime pair of extents, and its check. Scripts h420-h423 and h430-h432.
- docs/reports/2026-09-28-opus-scomp.md: the section on the signed moments. Scripts

h440–h448.

- docs/reports/2026-09-28-opus-twochain.md and 2026-09-28-opus-check8.md: Theorems 20 to 22, and their check. Scripts h450–h460 and h470–h476.
- docs/reports/2026-09-28-opus-dfour.md and 2026-09-28-opus-check9.md: Lemmas 13 to 20 and Theorem 23, and the check of them and of the section on the signed moments. Scripts h480–h485 and h530–h539.
- h550_lift.py: the check of Lemma 21.
- docs/reports/2026-09-28-opus-cubethree.md: the section on three chains in every pair of planes. Scripts h500–h518.
- docs/reports/2026-09-28-opus-sheet.md and 2026-09-29-opus-moment.md: the seventh item of the section on Theorem U. Scripts h560–h569 and h570–h579e.
- docs/reports/2026-09-30-fable-zeroflux.md and 2026-09-30-opus-check10.md: Lemmas 22 to 25 and the proof of Theorem 1 of 30 September, and their check. Scripts h600–h605 and h670–h672.
- docs/reports/2026-09-30-opus-kuwall.md: the first step of K_u beside a wall and its cone. Scripts h580–h589.
- docs/reports/2026-09-30-fable-rodline.md: Statement R, the box lemma and its certificates (certs/h633_R_b*, hashes in the report), the heights, and the integrality count. Scripts h630–h633. The zero-flux reach of (V) : h620_vflux.py.
- docs/reports/2026-10-01-opus-falsify.md: the counterexample search and the certificates of Theorem U' at sides seven and eight and on the four cells that are not cubic. Hashes and verdicts in certs/h74x_results.jsonl and certs/h74x_sha256_2026-10-01.txt; regenerate with h741_uprime.py --drat. Scripts h740–h744.
- docs/reports/2026-10-01-fable-screw.md and 2026-10-01-opus-rbox.md: the ninth item of the section on Theorem U. Scripts h700–h703 and h680–h688.
- docs/reports/2026-10-01-kucone-partial.md and 2026-10-01-opus-kucheck.md: the closed-form certificate of the first step of K_u and its check. Scripts h650–h656 and h720–h724.
- docs/reports/2026-10-01-opus-pacheck.md and 2026-10-01-opus-p1check.md: the checks of Lemma L, Lemma A, Theorem P, Lemma 21 and the chain to periodicity, and of the two linear-algebra steps of the first step of Theorem P. Scripts h730–h737 and h750–h753.

- docs/theorem-h-2026-09-06.md: the working note, with K_u and Theorem P.
- Scripts h240–h249 (reduction), h270–h272 (its check), h290–h299 (line laws), h310–h319 (Theorem 9), h320–h326 (its check), all in experiments/08_frozen_structure/scratch_h/.

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